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An advanced artificial intelligence framework integrating ensembled convolutional neural networks and Vision Transformers for precise soil classification with adaptive fuzzy logic-based crop recommendations

Farhan Sheth [a], Priya Mathur [b], Amit Kumar Gupta [a,*], Sandeep Chaurasia [a]

[a] Manipal University Jaipur, Jaipur-Ajmer Express Highway, Dehmi Kalan, Near GVK Toll Plaza, Jaipur, Rajasthan, 303007, India
[b] Poornima Institute of Engineering & Technology, ISI - 2, Poornima Marg, Sitapura, Jaipur, Rajasthan, 302022, India

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ABSTRACT

This study introduces an advanced Artificial Intelligence (AI) framework for soil classification and crop recommendation, combining Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) in an ensemble approach, alongside an adaptive fuzzy logic-based decision system for crop suggestions. While existing research typically addresses soil classification or crop recommendation in isolation, this work integrates cutting-edge deep learning models and fuzzy logic to enhance both tasks. The methodology is divided into two phases: Phase 1 covers data collection, preprocessing, and augmentation using Cycle Generative Adversarial Networks (CycleGAN) to expand the curated dataset of 1189 soil images to 8,413, while Phase 2 focuses on training CNN and ViT models, ensembling these models, and developing a fuzzy logic system that considers soil type, nutrients, potential of hydrogen (pH), and climatic conditions for crop recommendations. Experimental results indicate models achieve classification accuracies of up to 89.32 % on the original dataset, improving to 91.01 % with augmented data. On the CycleGAN-augmented (CyAUG) dataset, EfficientNet v2 Large and ViT-Large/16 attain accuracies of 99.60 % and 99.73 %, respectively. Furthermore, an ensemble of these architectures achieves a perfect accuracy of 100 %. The results are also validated by K-fold cross-validation. The research also presents 'Agro Companion,' an AI-powered tool that assists farmers in soil identification and crop selection based on geological and environmental data. This framework addresses key agricultural challenges in India, offering a high-accuracy, practical solution for improving both soil classification and crop recommendation. This research delivers state-of-the-art soil classification performance and a robust AI-based crop recommendation tool to support sustainable agricultural practices.

1. Introduction

Agriculture is the backbone of India's economy, supporting a vast majority of its population. With over 50 % of India's workforce engaged in farming, the success of agricultural practices is crucial for the country's food security and rural livelihood (Gulati and Juneja, 2022). However, many Indian farmers rely on traditional methods and personal judgment when deciding which crops to plant, often without conducting a comprehensive analysis of key factors like soil type, pH levels, nutrient content, rainfall, and climate conditions (Ganguly et al., 2017). This leads to frequent crop failures, reduced yield, and even land degradation. Consequently, there is a growing need for technology-driven solutions that can provide accurate soil assessments and crop recommendations to support farmers in making informed decisions. To address this challenge, this research aims to develop an application that leverages deep learning and fuzzy logic to classify soil types and recommend suitable crops. The proposed framework called Soil-ViT-CNN, uses a mobile-friendly interface to capture soil images, which is then analyzed to classify the soil type. Based on the soil classification, combined with location data and other known environmental information, such as pH levels and rainfall, the system provides tailored crop recommendations. Given the widespread availability of mobile phones, this application can enable farmers to assess their soil conditions quickly and easily, significantly reducing the guesswork and helping to mitigate crop failures. By

* Corresponding author.
E-mail addresses: farhan.219310185@muj.manipal.edu (F. Sheth), priya.mathur@poornima.org (P. Mathur), amit.gupta@jaipur.manipal.edu (A.K. Gupta), sandeep.chaurasia@jaipur.manipal.edu (S. Chaurasia).

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0952-1976/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

improving crop selection, this technology has the potential to reduce land degradation and enhance agricultural productivity. Recent advancements in artificial intelligence and deep learning (DL) have shown great promise in addressing complex agricultural challenges. Deep learning models, particularly Convolutional Neural Networks, have been widely applied in image-based soil classification due to their ability to extract intricate features from visual data. Vision Transformers, which employ self-attention mechanisms to capture global dependencies in images, have emerged as a powerful alternative to CNNs, especially for tasks involving complex textures and patterns (Padmapriya and Sasilatha, 2023; Gyasi and Purushotham, 2023a; Bharathi et al., 2023; Escorcia et al., 2022; Li and Iskander, 2023). This study integrates the strengths of both architectures in an ensemble approach, combining the local feature extraction capability of CNNs with the global context awareness of ViTs to improve the accuracy of soil classification. Additionally, the crop recommendation system utilizes fuzzy logic, which is well-suited for handling uncertainty and variability in agricultural data. By incorporating fuzzy logic, the system adapts its recommendations based on the soil properties and environmental factors specific to each location. This hybrid framework of soil classification and crop recommendation provides a comprehensive solution for precision agriculture. By offering both soil classification and crop recommendation in a user-friendly mobile application, this research aims to empower farmers with the tools needed to make data-driven decisions. This dual-purpose system not only enhances soil analysis but also provides actionable insights for sustainable crop management, contributing to a more productive and resilient agricultural sector in India.

In this study, several key contributions to advance soil classification and crop recommendation research are presented. Firstly, a robust, wide-ranging soil image dataset encompassing seven major soil types prevalent in India, Alluvial Soil, Black Soil, Laterite Soil, Red Soil, Yellow Soil, Arid Soil, and Mountain Soil is curated. While specifically tailored to Indian geographical conditions, the dataset is broadly applicable to regions worldwide with comparable soil characteristics. Furthermore, in light of the scarcity of diverse soil images, Generative Adversarial Networks, specifically, CycleGAN was utilized to augment the dataset. This process not only expands the volume of training examples but also heightens model robustness by introducing a greater variety of soil image characteristics and visual nuances.

To improve classification accuracy and fully leverage the varying strengths of modern computer vision architectures, the study develop hybrid ensemble models that combine cutting-edge CNNs and ViTs. By harnessing the complementary features of these two paradigms, the ensemble approach demonstrates enhanced performance and generalization capabilities. Beyond the image-centric focus, a rich repository of agricultural data, including information on soil nutrients, pH levels, climate indicators, and other key environmental parameters was also curated. These data points, extracted from both governmental databases and online sources form the backbone of a sophisticated crop recommendation system. The crop recommendation system accounts for nine essential input parameters, emphasizing the importance of soil-specific attributes (soil type, pH, nitrogen, phosphorus, and potassium content) while also integrating environmental factors such as rainfall, temperature, and humidity, alongside seasonal considerations. This comprehensive approach ensures that the recommendations can adapt effectively to varied climatic and agronomic conditions, thereby boosting the system's reliability across diverse scenarios. Notably, the recommendation engine is designed to support 23 different agricultural crops, providing farmers, agronomists, and researchers with nuanced insights tailored to the unique characteristics of the soil and the broader environment. Finally, to facilitate real-world adoption and deliver practical utility, a user-friendly web-based application that streamlines access to our system's recommendations was also developed. This intuitive interface enables stakeholders ranging from individual farmers to large-scale agricultural enterprises to make data-driven decisions, ranging from soil identification to crop selection. In sum, the research integrates a specialized soil image dataset, advanced augmentation techniques, hybrid deep learning models, and a rich agricultural information base to deliver a comprehensive and high-impact solution for sustainable, precision-driven agriculture.

2. Literature review

The use of artificial intelligence and deep learning techniques for soil classification has gained substantial momentum in recent years. This section reviews key literature on deep learning-based soil classification methods and crop recommendation systems, highlighting advancements and gaps that inform the development of the Soil-ViT-CNN framework.

2.1. Deep learning for soil classification

Several studies have employed deep learning algorithms for soil classification to improve accuracy and reduce the reliance on traditional, labor-intensive methods. Padmapriya and Sasilatha (2023) proposed a multi-labeled soil classification system that combines deep learning with empirical estimation techniques. Their approach highlights the critical role of accurate soil classification for sustainable agriculture, particularly in regions like India, where soil characteristics vary significantly across different terrains (Padmapriya and Sasilatha, 2023). Gyasi and Purushotham (2023) presented an in-depth analysis of current deep learning techniques for soil classification. They emphasized the strengths and limitations of various algorithms, noting the emerging trends of convolutional neural networks and transformer models in processing complex soil image data. CNNs have been widely adopted due to their superior ability to capture spatial features in images, making them a popular choice for soil classification tasks (Gyasi and Purushotham, 2023a). For instance, Bharathi et al. (2023) developed a hybrid deep learning model using the bat optimization algorithm and hyperspectral features for soil classification. Their approach demonstrated significant improvements in accuracy by optimizing the feature selection process (Bharathi et al., 2023). Similarly, Escorcia--Gutierrez et al. (2022) employed ensemble deep learning techniques to model soil nutrients and pH classification. Their model combined multiple classifiers to improve the robustness of predictions (Escorcia et al., 2022). Recent work by Li and Iskander (2023) focused on classifying sand using deep learning, illustrating the adaptability of these models across different soil types. They developed a CNN-based model specifically for sand classification, showcasing its versatility in geotechnical applications (Li and Iskander, 2023). In another study, Alsolai et al. (2023) leveraged a gradient-based optimizer in conjunction with deep learning to automate soil classification. Their model introduced an efficient training process that minimized computation time without compromising accuracy (Alsolai et al., 2023). Pandiri et al. (2024) developed a lightweight CNN model tailored for mobile applications, underscoring the importance of real-time soil classification systems. Their work aligns with the goals of the Soil-ViT-CNN framework, which aims to provide farmers with a mobile-friendly tool for soil analysis and crop recommendations (Pandiri et al., 2024). Similarly, Zhan et al. (2023) used deep learning in combination with the TDR cone penetration test to classify excavated soils, further expanding the application of AI in soil classification (Zhan et al., 2023). Qu et al. (2023) proposed an optimized Mask Region-based Convolutional Neural Network (Mask R-CNN) model for identifying uniformity in gravelly soil. Their work enhances the detection and classification of gravel particles, improving soil analysis in geotechnical applications (Qu et al., 2023). Kaur et al. (2023) developed a model that fuses multivariate time series meteorological and static soil data to predict crop yield using a multi-head self-attention network. This approach improves the accuracy of yield forecasts by integrating dynamic environmental factors (Kaur et al., 2023). Silva et al. (2023) explored the impact of soiling on photovoltaic panel efficiency using supervised learning techniques. Their study provides a data-driven approach to optimizing solar energy generation by

predicting the effects of dirt and dust on panel performance (Silva et al., 2023).

2.2. Ensemble methods and Vision Transformers

Recent advancements have seen the integration of multiple deep learning models to improve classification accuracy. Ensemble learning methods, which combine the predictions of multiple models, have proven effective in enhancing the robustness and performance of soil classification systems. For instance, Escorcia-Gutierrez et al. (2022) demonstrated the use of ensemble deep learning techniques to classify soil nutrients and pH levels. Vision Transformers have recently emerged as a powerful alternative to CNNs, particularly for tasks that require capturing global dependencies in image data (Escorcia et al., 2022). Gyasi and Purushotham (2023) explored the potential of ViTs in soil classification, noting their ability to outperform traditional CNNs in cases where soil textures and patterns are complex and not easily captured by local feature extraction alone (Gyasi and Purushotham, 2023b). Zhu et al. (2024) proposed a Semi-Supervised Classification and Stacking Learning (SSC-SL) approach, demonstrating the effectiveness of ensemble methods in soil classification, especially when labeled data is scarce (Zhu et al., 2024). Similarly, Kuan et al. (2025) provided a systematic review of machine learning and computer vision applications in agriculture, emphasizing deep learning models like CNNs and ViTs, along with data augmentation techniques such as GANs, to enhance classification accuracy (Ambareesh et al., 2025). Meanwhile, Ambareesh et al. (2025) explored the integration of ensemble learning and optimal type-2 fuzzy logic in Wireless Sensor Networks (WSNs), showcasing the ability of fuzzy logic to handle uncertainties in decision-making (Kuan et al., 2025). Together, these studies validate proposed approach of employing CNN-ViT ensembling for soil classification and incorporating fuzzy logic for crop recommendation. By combining these techniques, this research aims to develop a robust, data-driven system tailored to agricultural decision-making under uncertain conditions. The integration of both CNNs and ViTs, as proposed in the Soil-ViT-CNN framework, seeks to harness the strengths of both architectures—CNNs for local feature extraction and ViTs for global attention.

2.3. Fuzzy logic for crop recommendation

In addition to soil classification, several studies have explored crop recommendation systems that leverage fuzzy logic. Fuzzy logic systems are well-suited for agricultural applications, where uncertainty and variability in environmental factors often require adaptable decision-making frameworks. Hegde and Hegde (2023) proposed a fuzzy logic-based approach for soil classification and crop recommendation, highlighting its potential in reducing crop failure rates by considering factors such as soil type, pH, and climate (Hegde and Hegde, 2023). Rahmani et al. (2020) applied fuzzy logic in digital soil mapping, integrating it with machine learning to improve the accuracy of soil and land-use predictions (Rahmani et al., 2020). Shi et al. (2023) introduced a robust fuzzy ensemble framework using spectral learning for random projection-based fuzzy c-means clustering, demonstrating improved clustering performance under uncertainty (Shi et al., 2023). Similarly, Wang et al. (2024) applied fuzzy logic in environmental risk assessment, specifically for modeling soil heavy metal pollution, reinforcing its utility in handling imprecise and uncertain data (Wang et al., 2024). Expanding on this, Ramírez et al. (2024) proposed a hybrid approach combining general type-2 fuzzy systems and neural networks for clustering, classification, and world development indicator predictions, highlighting the synergy between fuzzy logic and deep learning models (Ramírez et al., 2024). Further advancements in fuzzy logic are explored by Castillo and Melin (2024), who introduced type-3 fuzzy logic for time series prediction, including a hybrid interval type-3 fuzzy-fractal approach, which enhances predictive accuracy by capturing higher levels of uncertainty (Castillo and Melin, 2024a, 2024b). Additionally, Melin et al. (2024) investigated the application of artificial neural networks in clustering, classification, and time series prediction, further validating the role of deep learning in intelligent decision-making systems (Melin et al., 2024). These studies collectively support presented research methodology, which integrates CNN-ViT ensemble learning for soil classification and fuzzy logic-based recommendation for crop selection, ensuring a robust and adaptable system for agricultural decision-making. This approach aligns with the objectives of the proposed Soil-ViT-CNN framework, which combines deep learning for soil classification with fuzzy logic for crop recommendations based on the classified soil properties.

2.4. Mobile-friendly applications and precision agriculture

The growing availability of mobile devices has opened new avenues for deploying AI-driven tools in agriculture. Several studies have focused on developing lightweight, mobile-friendly applications for real-time soil analysis. For example, Pandiri et al. (2024) developed a mobile-compatible soil classification system that allows users to capture soil images and receive instant classification results. This aligns with the goals of the proposed Soil-ViT-CNN framework, which is designed to be accessible via smartphones, providing farmers with an easy-to-use tool for soil analysis and crop recommendation (Pandiri et al., 2024). Inazumi et al. (2020) developed an AI system to support soil classification, further demonstrating the feasibility of mobile-based applications in precision agriculture (Inazumi et al., 2020). These advancements in mobile technology offer significant potential for improving agricultural practices, particularly in regions like India, where farmers have access to smartphones but may lack access to sophisticated agricultural technologies.

The existing literature underscores the growing importance of AI and deep learning in soil classification and crop recommendation. While CNNs remain the dominant choice for image-based classification tasks, ViTs are gaining traction for their ability to capture global dependencies in complex soil textures. Ensemble methods, which combine the strengths of multiple models, and fuzzy logic-based systems for crop recommendations offer a comprehensive solution for precision agriculture. The Soil-ViT-CNN framework aims to build on these advancements by integrating CNNs, ViTs, and fuzzy logic into a mobile-friendly platform, ultimately empowering farmers with data-driven tools to improve crop selection and enhance agricultural sustainability.

2.5. Motivation for the ensemble approach

The rationale for employing an ensemble of CNNs and ViTs in soil classification lies in the complementary strengths of these architectures. CNNs excel at capturing local spatial features, making them highly effective for texture-based classifications such as soil images. On the other hand, ViTs leverage self-attention mechanisms, allowing them to capture long-range dependencies and global contextual information. By integrating these two architectures through an ensemble approach, we aim to enhance the robustness and accuracy of soil classification, thereby mitigating the limitations of individual models and achieving a more generalized performance.

2.6. Motivation for the fuzzy logic-based recommendation system

The decision to integrate fuzzy logic into the crop recommendation system stems from its capability to handle uncertainty and imprecise data, which are inherent in agricultural decision-making. Traditional rule-based or machine learning approaches often struggle with ambiguous or overlapping conditions, such as variations in soil nutrient levels, pH values, and climatic factors. Fuzzy logic provides a flexible framework that can model expert knowledge and agricultural heuristics through linguistic rules, enabling more adaptive and interpretable

recommendations. This approach ensures that recommendations align closely with real-world agricultural practices, accommodating variations in soil conditions and environmental factors more effectively.

2.7. Research motivation

Soil classification and crop recommendation are crucial for optimizing agricultural productivity, yet existing methods suffer from key limitations, including.

- Limited datasets – Most available soil datasets are small and region-specific or lacks diversity, making it difficult to train robust machine learning models.
- Single-model dependency – Traditional approaches often rely on either CNNs or ViTs, which do not fully leverage the advantages of both architectures.
- Lack of Contextual Factors in Recommendations – Many existing crop recommendation systems rely on a narrow set of parameters: some base their suggestions solely on soil nutrients, pH, and climatic conditions while ignoring soil type, whereas others consider only soil type and overlook other key environmental factors. This narrow focus can significantly affect the accuracy of crop suitability assessments. In contrast, this study integrates all major, readily extractable factors relevant to crop recommendation, providing a more comprehensive and robust approach.
- Limited real-world applicability – Many soil classification models are not accessible to farmers, making it difficult to translate AI advancements into practical agricultural solutions.

To address these challenges, This study proposed Soil-ViT-CNN, an ensemble-based deep learning framework integrating ViTs and CNNs for accurate soil classification and a fuzzy logic-based crop recommendation system that incorporates multiple agricultural parameters.

3. Methodology

The objective of this research is to develop an advanced soil classification and crop recommendation system. The methodology is organized into two primary phases: Phase 1, focusing on data collection and preparation, and Phase 2, concentrating on system development. Phase 1 is further divided into two key components. The first component entails the acquisition and processing of soil and crop-related data from various online sources, while the second component focuses on data augmentation and preprocessing. Phase 2 comprises three essential tasks. The first task involves training machine learning models, specifically CNNs and ViTs, to construct accurate soil classification models. The second task is model ensembling, which combines the trained CNNs and ViTs with additional layers to enhance classification accuracy by leveraging their complementary strengths. The third task involves the creation of a fuzzy logic-based recommendation system. This task includes designing fuzzy logic rules and an inference system to recommend suitable crops based on classified soil types, as well as additional factors such as soil nutrient content, pH levels, and local weather conditions. The final step in Phase 2 involves deploying the fully integrated soil classification and crop recommendation system as a web-based application to ensure accessibility for end-users.

This study introduces several innovative methodologies, including the development of a comprehensive soil dataset tailored specifically to Indian geographical conditions, but with the potential for global applicability in regions with similar soil types. Additionally, the research explores the creation of hybrid models through the ensembling of cutting-edge CNNs and ViTs to maximize their individual advantages. Given the scarcity of diverse soil image datasets, GANs are employed for data augmentation to enhance the diversity and robustness of the training data. Moreover, extensive data, including information on soils, crops, climatic conditions, and key agricultural factors such as soil nutrients and pH levels, is collected from governmental and online sources to build a sophisticated crop recommendation system. This system is predominantly grounded in soil type but also accounts for various environmental and crop-specific factors. Lastly, a user-friendly web-based application is developed for practical, real-world use, allowing users to easily access the system's recommendations. The overall workflow of the methodology is illustrated in Fig. 1.

3.1. Dataset characterization

The study utilizes two primary datasets. The first is the **soil dataset**, comprising soil images used to train the deep learning models for soil classification. The second is the crop **recommendation dataset**, an extensive dataset that integrates multiple factors, including soil characteristics, crop information, nutrient levels, pH values, and climatic conditions. This comprehensive dataset, referred to as the recommendation dataset, serves as the core foundation for the crop recommendation system. In contrast, the dataset dedicated to soil image classification is referred to as the soil dataset.

3.1.1. Collection

For the soil dataset, soil images were acquired through a web scraping process facilitated by a Python script, which interacted with the Google image search engine. These images were gathered based on various user-uploaded entries under specific search queries. The search strategy employed keywords such as the specific soil name combined with terms like "soil' and "soil image' (e.g. Alluvial soil or Alluvial soil image) to ensure relevant and diverse image retrieval. In assembling the recommendation dataset, data was sourced from the official Indian government portal (https://data.gov.in) as well as from other reputable agricultural and plantation websites. This comprehensive data, encompassing a wide range of agricultural parameters, was meticulously consolidated to form a robust dataset designed to support accurate crop recommendations.

3.1.2. Soil dataset

India's agricultural landscape is characterized by diverse soil types, with this study focusing on seven primary classifications: **Alluvial Soil**, **Black Soil**, **Laterite Soil**, **Red Soil**, **Yellow Soil**, **Mountain Soil**, and **Arid Soil** (also referred to as Desert Soil). Each of these soil types plays a critical role in Indian agriculture. Fig. 2 provides a representative sample from each of these soil categories. After the soil images were collected via web scraping, the dataset underwent a rigorous manual cleaning process to ensure its relevance. Irrelevant images were discarded, and the remaining images were cropped to eliminate extraneous scenery. A significant challenge in compiling the soil image dataset was the scarcity of relevant images for certain soil types. While numerous images appeared in search results, many were irrelevant to the target soil class and subsequently discarded. This issue was particularly pronounced for Alluvial Soil and Yellow Soil, where finding adequate images was difficult. To ensure the dataset's quality and accuracy, an agricultural specialist verified the final collection, ensuring it was both proficient and reliable for use in developing an accurate soil classification model.

3.1.3. Recommendation dataset

The **recommendation dataset** was systematically organized into multiple segments following data collection. One segment focuses on crops, detailing their specific nutrient requirements, pH preferences, and ideal climatic conditions, such as rainfall, humidity, and temperature. Another segment incorporates historical, region-specific seasonal rainfall patterns, with data sourced from the Indian Meteorological Department. Additionally, a segment was dedicated to locality-specific temperature and humidity data. The final segment of the dataset focuses on soil, providing detailed information about each soil type's nutrient composition, pH levels, and the corresponding weather patterns

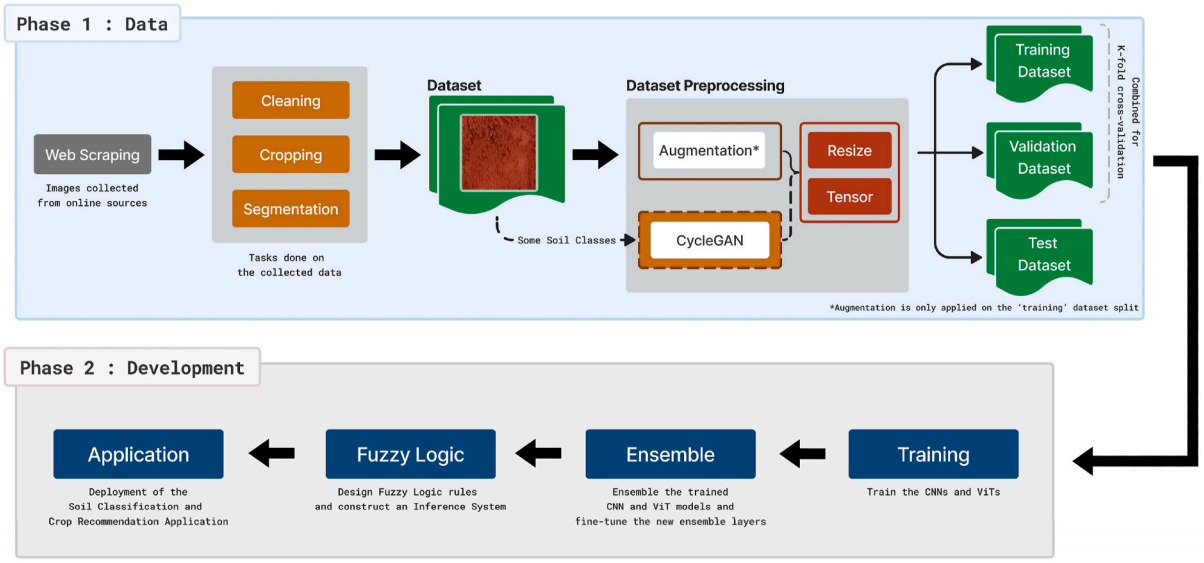


Fig. 1. The overview of the methodology employed in the study.

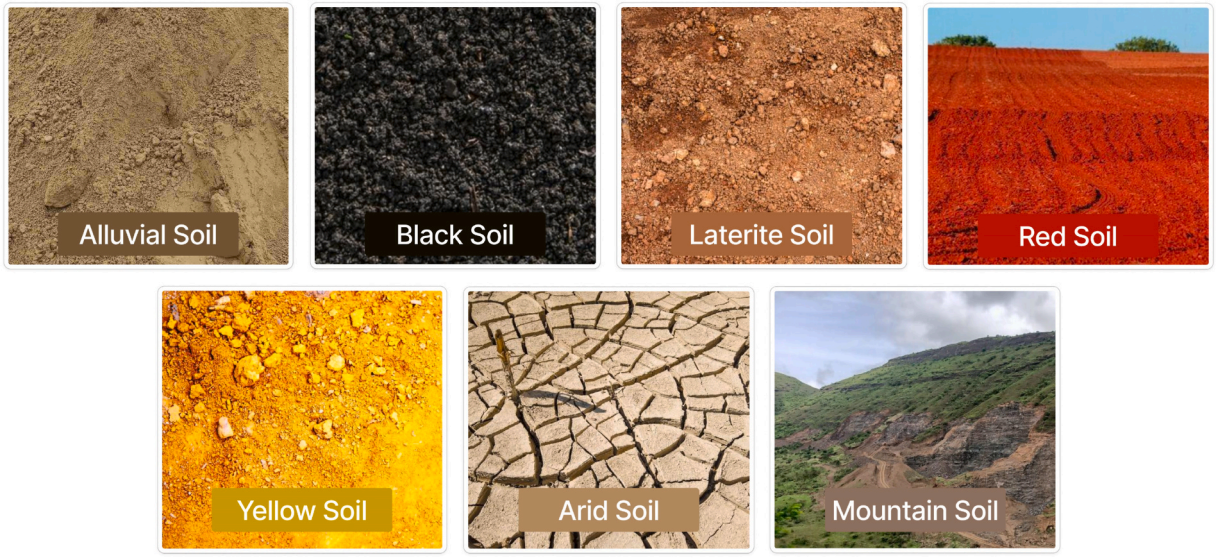


Fig. 2. Sample from each soil class.

typically associated with that soil. A total of nine input parameters were selected: **Soil Type**, **Season**, **pH**, **Rainfall**, **Temperature**, **Humidity**, **Nitrogen**, **Phosphorus**, and **Potassium** content. These input parameters were specifically chosen due to their availability and ease of access, as well as their critical role in determining crop growth and yield. Additionally, the dataset includes 23 crops as output parameters: Rice, Wheat, Maize (Corn), Cotton, Millets, Sugarcane, Rubber, Jute, Oilseeds, Pulses, Pearl Millet (Bajra), Lentil, Groundnut, Cocoa, Tea, Coffee, Flaxseed, Coconut, Oil Palm, Clove, Black Pepper, Cardamom, and Turmeric. Collectively, these segments form the recommendation dataset, which is used to develop tailored fuzzy logic rules and an inference system. This system enables precise crop recommendations based on soil type and various environmental and agricultural factors.

3.2. Dataset preprocessing

Preprocessing is a critical step in deep learning, particularly when dealing with datasets that may contain irregularities, noise, irrelevant information, or be of limited size. These challenges are often exacerbated in datasets obtained via web scraping, such as the one utilized in this study. Effective preprocessing mitigates these issues by removing inconsistencies and preparing the dataset for subsequent analysis, thereby improving its overall quality before being input into the models. Key preprocessing techniques include data augmentation, which can substantially increase the dataset's size and diversity. Additionally, deep learning models often have specific requirements related to image dimensions, quality, and format, and preprocessing ensures these criteria are met (Bhattacharyya, 2011). This preparatory phase is essential for enhancing the model's capacity to detect patterns and features, especially in the presence of dataset limitations. Ultimately, preprocessing contributes significantly to the model's robustness and its ability to generalize to unseen data. The specific preprocessing techniques applied in this study are outlined in the sections below.

3.2.1. Dataset augmentation

One of the primary limitations in developing effective neural

networks is the availability of sufficiently large datasets. Neural networks require extensive amounts of data to attain resilience, accuracy, and robustness (Van Der et al., 2014). A more substantial dataset can significantly improve model performance by providing a richer set of examples from which the model can learn. As a result, data augmentation has emerged as a vital preprocessing method. Data augmentation artificially increases the size and variety of the training split of the dataset by applying a range of transformations to the existing data. In this study, the training split of the soil dataset underwent various augmentation techniques. These techniques included random cropping, flipping, rotation, shearing, scaling, affine transformations, and adjustments to image quality through blurring, adding noise and shadows, and modifying brightness and contrast. By introducing these variations, data augmentation expands the diversity of the training dataset, thereby improving the model's ability to generalize and handle the complexities and nuances present in real-world data. Consequently, data augmentation is instrumental in boosting both the performance and reliability of the model.

3.2.2. *Augmentation using GAN*

While traditional data augmentation techniques can significantly enhance model performance, they have limitations. Although these methods increase the size of the training dataset, they do not necessarily improve the dataset's variability and randomness. Furthermore, in cases where the dataset is too small, applying excessive augmentation may lead to diminishing returns. This is particularly true for the collected soil dataset used in this study, where certain classes contain a very limited number of images due to a scarcity of available online data. In such cases, despite multiple augmentations, the underlying connections between the augmented images remain too similar, resulting in the model failing to learn distinctive patterns. Consequently, the model is at risk of overfitting on the training dataset, which may cause it to perform poorly when tested on unseen data.

To address this issue, a novel approach was adopted by generating synthetic images for the underrepresented soil classes. Generative Adversarial Networks, a powerful tool in modern machine learning, were employed to enhance the dataset. GANs are particularly effective for learning deep representations without the need for extensively labeled training data (Creswell et al., 2018). Specifically, this study utilized CycleGAN, a type of GAN designed to capture the characteristics of two image domains and learn how to translate these characteristics without requiring paired datasets (Zhu et al., 2017). CycleGAN employs two generators and two discriminators to overcome the limitations associated with paired data, making it a highly effective technique for tasks such as image domain transfer and style transformation (Yoo et al., 2020a). Previous research has shown that CycleGAN can help improve deep learning models by generating diverse training examples, thereby enabling the model to learn more effective decision boundaries between classes. Other supervised GAN techniques, such as conditional GANs, were not employed in this study due to the lack of paired images between soil classes. By generating synthetic images for the minority soil classes, CycleGAN substantially improved the dataset's generalizability and helped mitigate class imbalance (Yoo et al., 2020b). Fig. 3 illustrates the architecture of the CycleGAN model used in this study.

In this study, synthetic images were generated for soil classes with limited availability, such as alluvial soil and yellow soil, as well as for larger classes like red soil and black soil. The selection of these particular soil classes was guided by several criteria. One key factor is that these soil types exhibit correlations with each other, which is essential for the effective use of CycleGAN. CycleGAN requires some degree of similarity between classes to enable feature transfer; if the classes are too dissimilar, the model would struggle to perform the image translation. For instance, soil types such as arid soil, laterite soil, and mountain soil exhibit substantial variations: arid soil is typically associated with deserts, mountain soil features highly variable land structures, and laterite soil often has a rocky texture. These distinct characteristics pose challenges for CycleGAN because even after translation, the synthetic images may retain distinguishing features of the original classes, making them unsuitable for augmentation (Sandfort et al., 2019). On the other hand, alluvial soil, red soil, yellow soil, and black soil share more similarities and are also among the most prevalent soil types in India. By generating synthetic images for these soil classes, the model gains access to a more diverse and representative dataset, thereby improving its robustness and resilience. Furthermore, this augmentation enhances the model's ability to generalize across a broader range of soil types.

Before applying CycleGAN, the dataset was further refined by removing images that exhibited extreme variations from the typical characteristics of their respective soil classes. For example, images that provided highly magnified views of soil textures, which were noticeably different from the standard images in that class, were excluded. Such images would be difficult to identify as part of a new class even after image translation, potentially undermining the model's accuracy. To

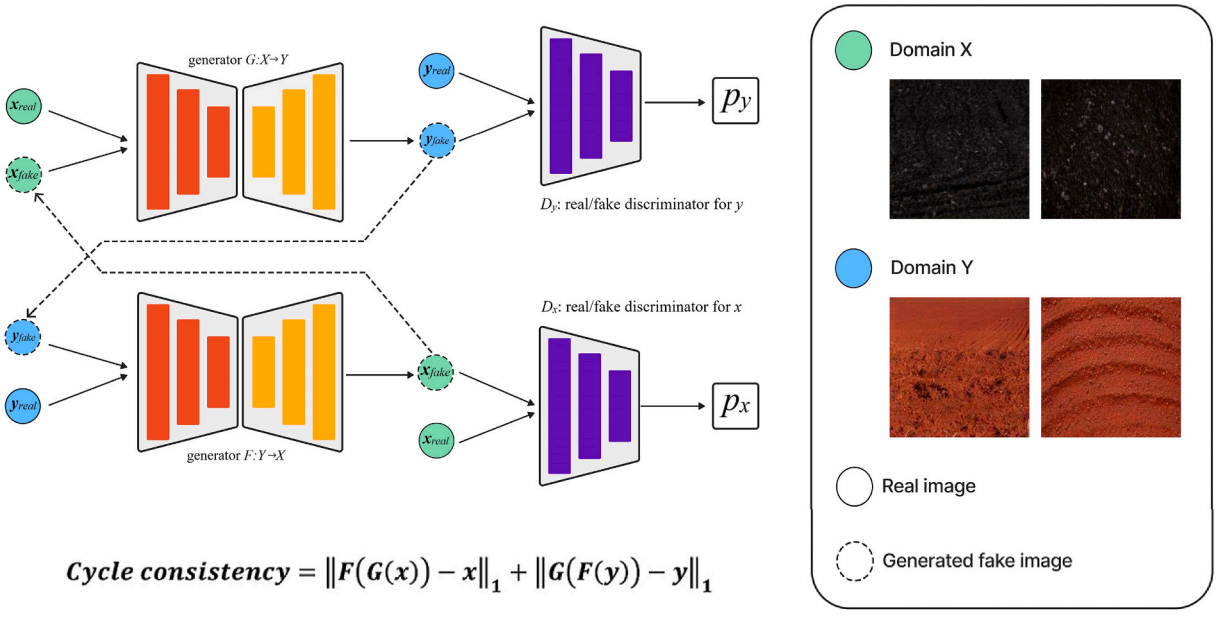


Fig. 3. Schematics of the CycleGAN model.

ensure optimal performance of the CycleGAN model and generate a large number of high-quality synthetic images, the training split of the pruned dataset was augmented. For the four selected soil classes, twelve internal transformations would be done, involving six pairs over which the CycleGAN models were trained. The relationship of image transformations over a set is expressed as follows in Equation (1):

$$\{AB : A \leftrightarrow B\}, \textit{where } A \rightarrow B \textit{ translation and } B \rightarrow A \textit{ translation} \quad (1)$$

This equation represents the bidirectional learning process of the model, where it learns to translate images from class A to class B and vice versa. Through this approach, both soil classes are augmented, enriching the dataset for both categories. The six transformation pairs used in this study are:

$\{\text{AY} : \textit{Alluvial Soil} \leftrightarrow \textit{Yellow Soil}\}$

$\{\text{BA} : \textit{Black Soil} \leftrightarrow \textit{Alluvial Soil}\}$

$\{\text{RA} : \textit{Red Soil} \leftrightarrow \textit{Alluvial Soil}\}$

$\{\text{BY} : \textit{Black Soil} \leftrightarrow \textit{Yellow Soil}$

$\{\text{BR} : \textit{Black Soil} \leftrightarrow \textit{Red Soil}\}$

$\{\text{RY} : \textit{Red Soil} \leftrightarrow \textit{Yellow Soil}\}$

This approach ensured a balanced augmentation process across the selected soil classes, thereby enhancing the dataset's variability and improving the model's overall generalization capabilities.

3.2.3. Image resizing

To ensure compatibility with the input size requirements of Convolutional Neural Network and Vision Transformer models, the soil dataset was resized while preserving its original content. The resizing was specifically tailored to meet the standard input dimensions required by each model type. CNN models typically utilized images resized to 256 × 256 pixels, while ViT models required 224 × 224 pixels. This resizing process facilitated consistency in input dimensions across all models, ensuring uniformity in training and testing. Additionally, it allowed the dataset to be aligned with standard image dimensions, enabling the model to generalize effectively to real-world data inputs.

3.2.4. Dataset splitting

The soil dataset was systematically partitioned into three subsets to optimize its utility for CNN and ViT models: 70 % of the data was designated for training, and 15 % each was allocated for validation and testing. This partitioning strategy was carefully designed to enable a comprehensive evaluation of the model's performance at different stages of development. The training subset, encompassing 70 % of the dataset, provided the foundational data from which the model could learn and adapt to the underlying patterns present in the soil imagery. The validation subset, comprising 15 % of the dataset, was used to monitor the model's performance during training, allowing for an assessment of its generalization capabilities. The remaining 15 % was reserved exclusively for testing, ensuring that the model's performance on unseen data could be rigorously evaluated—thereby offering an accurate measure of its real-world applicability.

For the CycleGAN models, a different approach was applied, with each set split into training and testing subsets. In this case, 85 % of the images were used for training, and the remaining 15 % for testing, enabling the evaluation of the model's ability to translate between different domains of soil imagery.

To mitigate the risk of overfitting, K-fold cross-validation was employed as a robust technique for model evaluation. In this approach, the combined training and validation dataset, which represented 85 % of the original data, was further divided into K subsets or folds. Iteratively, K-1 folds were used for training while the remaining fold was reserved for validation, rotating this process until each data point was used for validation exactly once. This method not only addressed overfitting—where models perform well on training data but struggles with unseen data—but also enhanced the robustness and reliability of the model's performance estimates across multiple subsets of the dataset. It is important to note that while K-fold cross-validation was applied to the training and validation sets, the original testing set, comprising 15 % of the data, remained untouched throughout the process. This preserved the integrity of the final evaluation, ensuring that the model's performance could be assessed on a genuinely independent dataset. By employing this method, the study produced a more accurate and impartial measure of model performance across diverse data partitions, ultimately reinforcing the credibility of the evaluation results (Wong and Yeh, 2019).

3.3. Development of models

This study explores the development of various deep learning models, primarily focusing on Convolutional Neural Networks and Vision Transformers. The CNN models employed in this research include MobileNet v3 Large, ResNet101, ResNet152, and EfficientNet v2 Large, while the Vision Transformer models consist of ViT-B/16 and ViT-L/16. To enhance model performance, a transfer learning approach was utilized for training, allowing the models to leverage pre-trained weights and adapt more efficiently to the soil dataset. Additionally, ensemble learning techniques were applied, combining multiple models to create a more robust and accurate predictive framework. All models were rigorously evaluated using a comprehensive set of performance metrics, ensuring a thorough assessment of their effectiveness and precision in addressing the task at hand.

3.3.1. Convolutional neural network

In recent years, deep learning has achieved remarkable success across numerous application domains, with Convolutional Neural Networks playing a pivotal role in this advancement. Since the publication of the groundbreaking AlexNet model (Alom et al., 2018), CNNs have experienced significant growth and have been extensively applied in image processing, along with many other complex tasks. CNNs are particularly distinguished by their use of convolutional layers, which are highly effective at detecting local patterns in images. These networks typically consist of four key types of layers: convolutional, max pooling, fully connected, and output layers, which are arranged in a sequential manner. One of the most notable strengths of CNNs is their architectural flexibility, which allows for customizations tailored to specific tasks. This adaptability is evident in CNNs' dual role as both feature extractors and classifiers. Mathematically the feature value at location (i, j) in the kth feature map of lth layer, $z^l_{i,j,k}$ is calculated by Equation (2) (Li et al., 2021).

$$z^l_{i,j,k} = {w^l_k}^T x^l_{i,j} + b^l_k \quad (2)$$

Here, $x^l_{i,j}$ is the input patch from the previous layer, w^l_k denoted the weight vector associated with the kth filter in the lth layer, and b^l_k represents the corresponding bias term. Through backpropagation, the weights w^l_k are iteratively adjusted in response to classification errors.

Let $a(.)$ denotes nonlinear activation function. The activation value $a^l_{i,j,k}$ of convolution feature $z^l_{i,j,k}$ can be calculated as:

$$a^l_{i,j,k} = a\left(z^l_{i,j,k}\right) \quad (3)$$

Let pool(.) denoting the pooling function and $\mathscr{R}_{i,j}$ is local neighborhood around location (i, j). Then the pooling function $y^l_{i,j,k}$ for each feature map $a^l_{:,:,k}$ defined as:

$$y^l_{i,j,k} = pool\left(a^l_{m,n,k}\right), \forall (m,n) \in \mathscr{R}_{i,j} \quad (4)$$

The convolutional and pooling layers in CNNs extract intricate features from the input data, while the fully connected layers are responsible for classifying this data based on the learned high-level features. CNNs utilize a variety of activation functions to model complex patterns, and there is a wide range of CNN architectures that vary in depth, width, and complexity (Li et al., 2021). In this study, the CNN models employed include Residual Neural Network101 (ResNet101) and Residual Neural Network152 (ResNet152) from the ResNet family, renowned for their deep architectures and use of residual connections. These connections enable the models to learn complex features effectively while mitigating the vanishing gradient problem, which often hampers the training of deep networks. The ResNet models are highly effective at capturing intricate patterns in data (He et al., 2016). Additionally, Mobile Neural Network Version3 (MobileNetV3) Large was utilized due to its efficiency and lightweight design. Its depth wise separable convolutions reduce the model size and computational requirements, resulting in faster inference while maintaining robust performance (Howard et al., 2019). Lastly, EfficientNetV2 Large was selected for its balanced optimization of accuracy and efficiency. This model scales networks across depth, width, and resolution in a balanced manner, achieving high accuracy with fewer parameters and lower computational costs (Tan and Le, 2021). Fig. 4 (A) illustrates the fundamental structure of CNN models. These architectures provide a strong foundation for the classification tasks performed in this study, making them highly suitable for the diverse challenges addressed in the research.

3.3.2. Vision Transformers

The success of Transformer models in natural language processing has sparked significant interest within the vision research community to explore their potential in addressing computer vision challenges. Transformers offer several notable advantages, particularly in their ability to model long-range dependencies between elements in an input sequence, while also supporting parallel processing, unlike recurrent neural networks. Furthermore, these models are versatile, enabling the processing of multiple modalities such as images, videos, text, and speech using a uniform architecture. This adaptability, combined with their ability to scale efficiently to very large networks and extensive datasets, has been a key factor in their growing popularity across various domains. A major reason for the success of Transformers, particularly in vision tasks, is the self-attention mechanism. This mechanism allows for the effective capture of long-term dependencies between sequence elements, an area where conventional recurrent models, such as Recurrent Neural Network (RNN) and Long Short-Term Memory (LSTM), often struggle. In contrast, self-attention excels at encoding these relationships, leading to superior performance in many tasks (Khan et al., 2022). Equation (5) shows the output Z of self-attention layer in transformers. Let denote a sequence of n entities $((x_1, x_2, \ldots\ldots x_n)$ by $X \in \mathbb{R}^{n \times d}$ where d is the embedding dimension to represent each entity. The goal of self-attention is to capture the interaction amongst all n entities by encoding each entity in terms of global contextual information. This is done by defining three learnable weight matrices to transform the

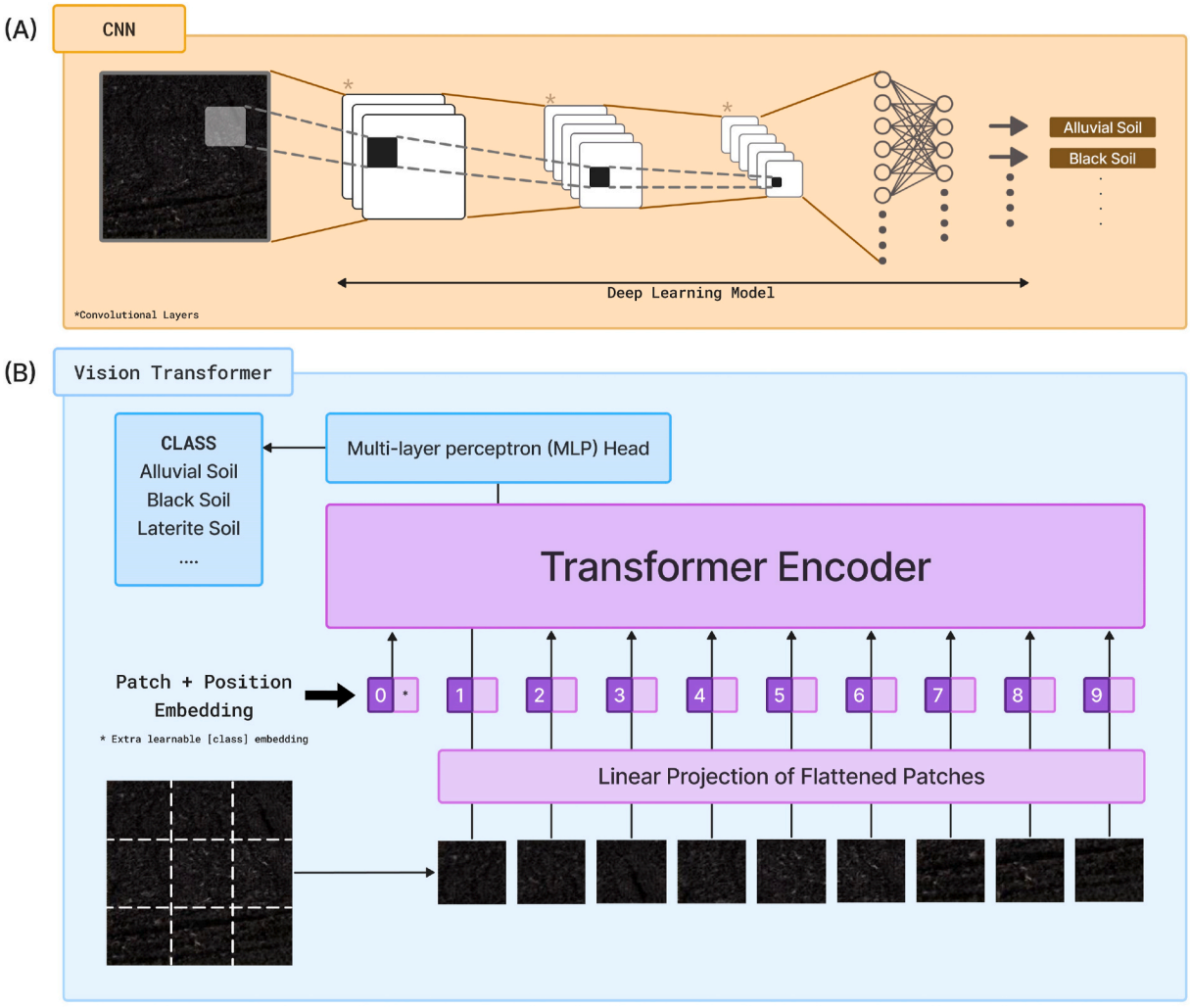


Fig. 4. Overall structure of CNN models (A) and ViT models (B).

Queries $\left(W^Q \in \mathbb{R}^{d\times d_q}\right)$, keys $\left(W^K \in \mathbb{R}^{d\times d_k}\right)$, and Values $\left(W^V \in \mathbb{R}^{d\times d_v}\right)$, where $d_q = d_k$. The input sequence X is first projected onto these weight matrices to get $Q = XW^Q$, $K = XW^K$, and $V = XW^V$. The output $\left(Z \in \mathbb{R}^{d\times d_v}\right)$ of the self-attention layer is,

$$z = softmax\left(\frac{QK^T}{\sqrt{d_q}}\right)V \tag{5}$$

The Vision Transformer and its subsequent variations have demonstrated that self-attention-based architectures can surpass traditional Convolutional Neural Networks in a wide range of vision applications (Mao et al., 2022). In this study, two variants of the ViT model—ViT-Base/16 (ViT-B/16) and ViT-Large/16 (ViT-L/16)—were employed. These models differ primarily in their size and computational complexity. ViT-B/16 consists of 12 transformer encoder layers, making it more computationally efficient, while ViT-L/16 comprises 24 transformer encoder layers, resulting in greater performance but at the expense of higher computational demands. The structure of the Vision Transformer models is illustrated in Fig. 4 (B). ViT-B/16 is particularly suited for scenarios where computational resources are limited, whereas ViT-L/16 offers enhanced performance for tasks requiring greater model capacity and precision.

3.3.3. Transfer learning

To enable accurate identification of various soil types, it is essential to train models on a custom dataset. However, creating and training large models on relatively small datasets poses significant challenges, particularly in terms of computational resource constraints and the risk of overfitting (Tajbakhsh et al., 2016). To mitigate these issues, a technique called transfer learning is often applied. Transfer learning leverages models that have been pre-trained on larger datasets, such as the ImageNet dataset, and fine-tunes them on a smaller, custom dataset. This approach allows the pre-trained models' extensive knowledge to be transferred and adapted for the smaller dataset (Tan et al., 2018). In this study, both CNN and ViT models were initialized using their ImageNet pre-trained weights. The final layer, responsible for classifying ImageNet classes, was replaced with a new fully connected layer tailored to the specific number of soil types in the custom dataset. A variant of transfer learning known as fine-tuning was employed. Unlike traditional transfer learning, where only the newly added layers are trained, fine-tuning allows a larger portion or even the entire set of model parameters to be updated. This method has demonstrated greater effectiveness when the custom dataset significantly differs from the original dataset used for pre-training, such as ImageNet (Sharif et al., 2014).

For fine-tuning the CNN models, the Adaptive Moment Estimation (Adam) optimizer was utilized. Adam is a first-order gradient-based optimization algorithm that is particularly effective in both online and non-stationary settings. It combines the strengths of Adaptive Gradient Algorithm (AdaGrad), and Root Mean Squared Propagation (RMSProp) and performs well with sparse gradients (Kingma, 2014). For the ViT models, a variant of Adam known as AdamW was selected. AdamW has been shown to outperform Adam in training transformer-based models by decoupling weight decay from the gradient updates, leading to better regularization and improved training dynamics (Loshchilov and Hutter, 2017). This separation enhances model generalization by mitigating the risk of overfitting during fine-tuning.

Two critical hyperparameters that influence training outcomes are the learning rate and weight decay. These hyperparameters control the step size during optimization and play a key role in ensuring efficient model convergence. To identify the optimal values for these hyperparameters, a Grid Search method was employed. Grid Search systematically explores various combinations of predefined hyperparameter values, ensuring the selection of the best combination for the given dataset and model architecture. Given the relatively small dataset size and the use of lighter models, this method was computationally feasible (Ahamad et al., 2023). The careful tuning of these hyperparameters ensured the models were trained both efficiently and effectively, maximizing the potential of the chosen optimization algorithms. The mathematical formulation of cross-entropy loss is detailed in Equations (6) and (7).

$$L_{cross\ entropy} = -\sum_{i}^{N} y_i \log(p_i) \tag{6}$$

$$L_{cross\ entropy\ for\ multi\ class} = -\frac{1}{N}\sum_{i}^{N}\sum_{j}^{M} y_{i,j} \log\left(p_{i,j}\right) \tag{7}$$

Where y_i, $y_{i,j}$ represents the true values. p_i represents the probability value predicted by the classifier for the ith image. $p_{i,j}$ represents the probability assigned by the classifier that sample i belongs to class j. N denotes total number of rows and M represents the total number of classes in the dataset (Ahamad et al., 2023). The optimizer iteratively updates the model's parameters to minimize the loss function, ultimately improving model performance over time.

3.3.4. Ensembling

After fine-tuning the Convolutional Neural Network and Vision Transformer models on the soil dataset, they were combined using an ensemble technique. Ensemble models that integrate CNNs and ViTs harness the strengths of both architectures, enhancing overall performance, especially in tasks where misclassification can have costly consequences, such as in agriculture. In such scenarios, incorrect predictions could lead to the misallocation of resources. CNNs are adept at capturing local features through hierarchical filtering, making them particularly powerful for tasks such as object detection and segmentation. Examples include models like ResNet, which excels at detecting intricate patterns; MobileNets, known for depth wise separable convolutions that make them faster yet reliable; and EfficientNet, which scales networks in a balanced manner across depth, width, and resolution, ensuring both accuracy and efficiency. In contrast, ViTs employ self-attention mechanisms to capture global context and long-range dependencies, which is particularly beneficial for complex tasks like image classification (Van Der et al., 2014; Creswell et al., 2018; Zhu et al., 2017; Yoo et al., 2020a, 2020b). By ensembling CNNs and ViTs, a more robust and accurate system is achieved, benefiting from CNNs' ability to extract detailed local features and ViTs' capacity for understanding global contexts. This synergy improves generalization and enhances the model's resilience to varying data distributions, leading to superior performance overall (Jiang et al., 2021).

In the ensembled architecture, the previously added fully connected layers for soil classification were removed. Instead, a series of layers were introduced. Features extracted from the CNN and ViT models' earlier layers were concatenated and fed into a fully connected layer with an output of 512 features. These outputs were then passed through a Rectified Linear Unit (ReLU) activation layer, which aids the model in learning non-linear and complex patterns (Schmidt- and Hieber, 2020). The mathematical formulation of the ReLU function is provided in Equation (8):

$$R(x) = \max(0, x) \tag{8}$$

The features from the ReLU layer were then input into another fully connected layer, which reduced the 512 features to 7, corresponding to the number of soil types in the dataset. Finally, the output of this layer was passed through a softmax function, which converts the neural network's raw outputs into a probability distribution, making the results interpretable (Li et al., 2022). The softmax function is mathematically defined in Equation (9):

$$softmax(\boldsymbol{z})_i = \frac{e^{z_i}}{\sum_{j=1}^{N} e^{z_j}} \tag{9}$$

where z_i denoted the raw output from the neural networks, e^{z_i} is the

exponential function of the input vector, e^{z_j} is the exponential function of the output vector and N is the number of classes. The i-th entry in the softmax output vector softmax($\mathbf{z}$) can be thought of as the predicted probability of the test input belonging to class i. For all ensemble architectures, only the newly added final layers are fine-tuned on the dataset, while the already trained CNN and ViT layers remain frozen. Fig. 5 illustrates the structure of the ensembled model, which integrates both the CNN and ViT models.

The study incorporates eight different ensemble models, each designed to leverage the combined strengths of their constituent architectures to optimize performance, resilience, and robustness. The ensemble models are listed in Table 1. By combining the capabilities of various individual models, these ensembles provide enhanced overall performance.

One notable ensemble is the **Mob-VB**, which integrates MobileNet v3 Large and ViT-B/16. This combination strikes a balance between the lightweight efficiency of MobileNet and the rich feature representation capabilities of the Vision Transformer. The **R101-VB** ensemble, which combines ResNet101 and ViT-B/16, excels in deep feature extraction and global context understanding. It is well-suited for tasks requiring high accuracy, efficiency, and precise recognition of intricate details. These models are ideal for applications where model size and inference speed are critical, such as mobile or edge device deployment, but a high level of accuracy is also necessary.

For more complex agricultural tasks, ensembles like **R152-VL** (ResNet152 + ViT-L/16) and **EffL-VL** (EfficientNet v2 Large + ViT-L/16) offer robust solutions. The deep architecture of ResNet152 in the R152-VL ensemble allows for precise handling of complex images, while the EffL-VL ensemble balances power and efficiency, making it scalable for larger datasets. The **Mob-R101-VB** ensemble, which combines MobileNet with ResNet101 and ViT-B/16, offers versatility across various applications by combining the efficiency of MobileNet with the deep learning capabilities of ResNet101 and the global context awareness of ViT-B/16. These ensembles are particularly suitable for large-scale deployments in cloud environments, where top-tier accuracy is prioritized alongside a degree of computational efficiency.

The **R152-EffL-VL** ensemble (ResNet152 + EfficientNet v2 Large + ViT-L/16) is engineered to achieve state-of-the-art accuracy, especially in challenging tasks that require a balance between detailed feature extraction and computational efficiency. The ultimate ensembles—**Mob-R101-EffL-VB** (MobileNet v3 Large + ResNet101 + EfficientNet v2 Large + ViT-B/16) and **Mob-R152-EffL-VL** (MobileNet v3 Large + ResNet152 + EfficientNet v2 Large + ViT-L/16)—combine the strengths of nearly all cutting-edge CNNs and ViTs. These models are designed for a wide range of complex image classification tasks where both high proficiency and accuracy are crucial. They are particularly suited for advanced research applications and scenarios where maximum accuracy is essential, such as precision agriculture. Each ensemble model offers distinct advantages based on the application context, ranging from mobile and edge device deployment to high-resolution, complex image analysis, and large-scale AI systems (Vasan et al., 2020).

3.3.5. Model's evaluation

To thoroughly evaluate the performance of the models, they were subjected to extensive testing using a comprehensive set of evaluation metrics to ensure robust assessment. The key metrics calculated included Accuracy, Specificity, Precision, Sensitivity, and the F1-score (Kamilaris and Prenafeta-Boldú, 2018). These metrics provide a well-rounded understanding of the models' predictive capabilities, each capturing different dimensions of their effectiveness. Equations (10)–(14) present the mathematical representations of these metrics:

$$Accuracy = \frac{count\ of\ accurate\ predictions}{overall\ count\ of\ samples} \tag{10}$$

$$Specificity = \frac{TN}{TN + FP} \tag{11}$$

$$Precision = \frac{TP}{TP + FP} \tag{12}$$

$$Sensitivity = \frac{TP}{TP + FN} \tag{13}$$

$$F1 - score = \frac{2 \times \text{Precision} \times \text{Sensitivity}}{Precision + Sensitivity} \tag{14}$$

In these formulas, TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively (Dai and Berleant, 2019). By evaluating these metrics, a holistic view of model performance is achieved, ensuring that both positive and negative classifications are appropriately considered. In addition to these metrics, graphs showing the models' training accuracy and loss were generated to provide further insight into the learning trajectory during the training

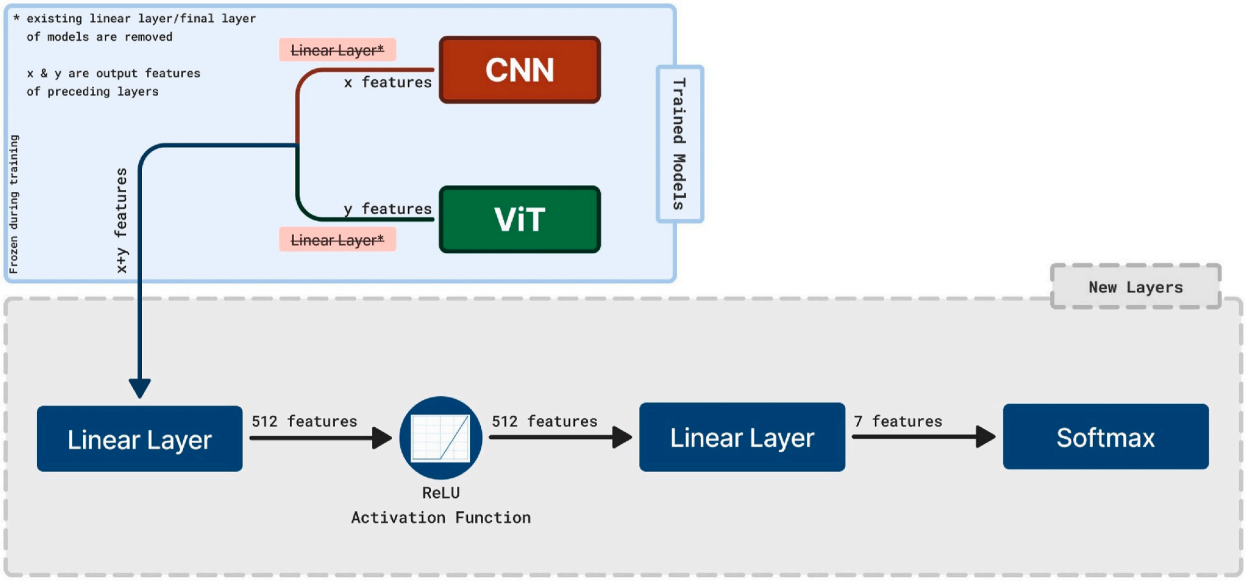


Fig. 5. The structure of ensembled CNN and ViT models.

Table 1
Structure of ensembled models utilized in the study.

Models included						*Ensembled Model*
Convolutional Neural Networks				Vision Transformers		
MobileNet v3 large	ResNet101	ResNet152	EfficientNet v2 large	ViT-B/16	ViT-L/16	
✓	×	×	×	✓	×	*Mob-VB*
×	✓	×	×	✓	×	*R101-VB*
×	×	✓	×	×	✓	*R152-VL*
×	×	×	✓	×	✓	*EffL-VL*
✓	✓	×	×	✓	×	*Mob-R101-VB*
×	×	✓	✓	×	✓	*R152-EffL-VL*
✓	✓	×	✓	✓	×	*Mob-R101-EffL-VB*
✓	×	✓	✓	×	✓	*Mob-R152-EffL-VL*

phase. These graphs offer a nuanced understanding of how the models improved over time and help identify potential overfitting or underfitting issues.

3.4. Recommendation system

The crop recommendation system is a crucial component of this study, designed using fuzzy logic. The fuzzy inference system, based on specific input factors, recommends an appropriate crop. Specifically, nine input factors are utilized to determine the optimal crop recommendation from a set of 23 crops. The recommendation dataset utilized for this system is thoroughly integrated into the design of the fuzzy logic model, ensuring a comprehensive and accurate crop selection process.

3.4.1. Fuzzy logic

Fuzzy logic is an extension of traditional Boolean logic, which deals with the concept of partial truth—values that exist between "completely true' and "completely false.' This flexibility allows for the representation of scenarios where truth values are not absolute, which is essential for addressing the inherent uncertainties found in real-world problems. Fuzzy logic operates on a continuum of values between 0 (representing completely false) and 1 (representing completely true) (Zadeh, 2008). This flexibility makes fuzzy logic particularly advantageous in handling imprecise or uncertain data that cannot be accurately represented using binary, crisp sets. In a fuzzy set, elements are paired with membership functions that define their degree of belonging to a particular set. The membership value ranges from 0 to 1, where 0 indicates the lowest degree of membership and 1 represents the highest (Mendel, 1995). Fig. 6 provides an overview of the fuzzy inference system employed in this study to recommend crops. By allowing partial truth values, fuzzy logic enhances the recommendation system's ability to account for real-world complexities and uncertainties in crop selection.

3.4.2. Fuzzy based recommendation system

A fuzzy inference system comprises three key stages: fuzzification, inference, and defuzzification. During fuzzification, crisp input values are converted into fuzzy set values using fuzzy techniques. This process allows the system to handle imprecise and uncertain data. The inference system then utilizes membership functions to analyze the relationships between these inputs. The degree of fuzziness in the system is represented by the membership functions, which quantify the degree of truth and are typically depicted in graphical forms. These visual representations facilitate the interpretation of fuzzy set values for the input parameters (Ahmadi et al., 2020). Common membership functions used in fuzzy logic include singleton, triangular, trapezoidal, Gaussian, generalized bell-shaped, and sigmoidal functions (Ali et al., 2015). In this study, two widely adopted membership functions—triangular (trimf) and trapezoidal (trapmf) were implemented, to model the recommendation dataset. These functions were chosen for their optimal balance of interpretability, simplicity, and computational efficiency. Among various candidate membership functions such as singleton, Gaussian, generalized bell-shaped, and sigmoidal, the triangular and trapezoidal functions demonstrate superior alignment with the underlying characteristics and distribution of the data. Moreover, these functions facilitate intuitive interpretation and tuning of parameters, essential in recommendation scenarios. The mathematical formulations of the triangular and trapezoidal functions are presented in Equations (15)–(17).

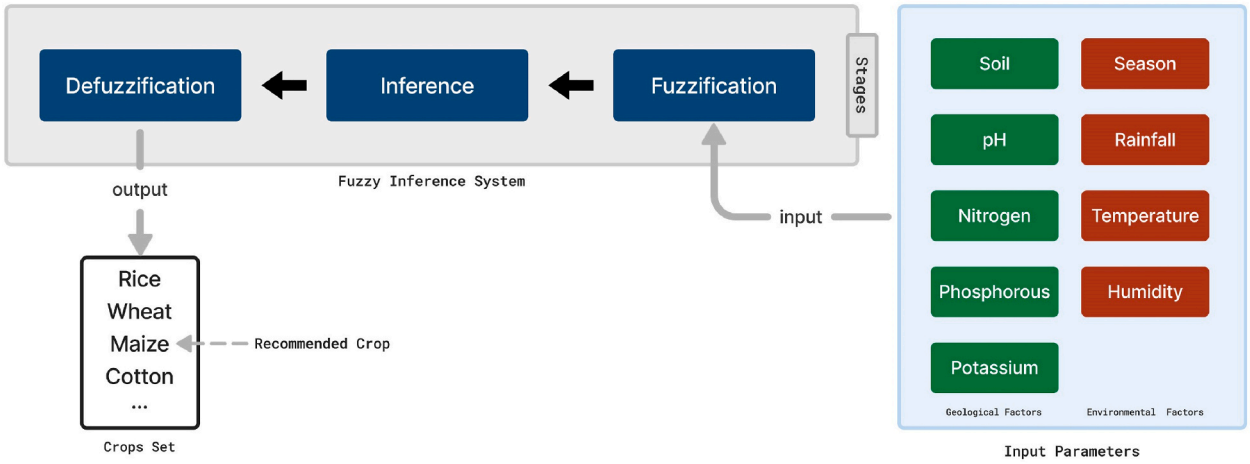


Fig. 6. Workflow of the fuzzy inference system employed in the study.

$$\mu_{trimf}(x;a,b,c) = \begin{cases} 0, & x \le a \\ \dfrac{x-a}{b-a}, & a \le x \le b \\ \dfrac{c-x}{c-b}, & b \le x \le c \\ 0, & x \ge c \end{cases} \tag{15}$$

$$\mu_{trampf}(x;a,b,c,d) = \begin{cases} 0, & x \le a \\ \dfrac{x-a}{b-a}, & a \le x \le b \\ 1, & b \le x \le c \\ \dfrac{d-x}{d-c}, & c \le x \le d \\ 0, & x \ge d \end{cases} \tag{16}$$

$$\mu_{trapezoid} = \max\left(min\left(\frac{x-a}{b-a}, 1, \frac{d-x}{d-c}\right), 0\right) \tag{17}$$

The fuzzy-based recommendation system uses nine different input parameters to recommend crops, which are divided into geological and environmental factors. The geological factors include Soil type, pH, and the content of Nitrogen (N), Phosphorus (P), and Potassium (K) in the soil. The environmental factors are Season, Rainfall, Temperature, and Humidity. The output of the system is a recommended crop based on these input factors. These specific parameters were selected due to their significant role in crop selection and growth, as well as their availability in real-world agricultural settings.

Each of these geological and environmental factors plays a crucial role in the crop recommendation system. Soil type is fundamental, as different crops thrive in particular soil textures, which promote optimal root development and nutrient absorption. In this study, soil type is determined through deep learning models that analyze soil images. Soil pH is also important, as it influences the availability of nutrients; some crops prefer acidic soils, while others perform best in neutral pH conditions. The soil's Nitrogen, Phosphorus, and Potassium (NPK) content has a direct impact on plant health and growth: nitrogen promotes leaf development, phosphorus supports root formation, and potassium enhances the plant's resistance to stress. On the environmental side, factors such as Season determine optimal planting times and crop growth cycles. Rainfall is critical for ensuring sufficient water supply, with each crop having specific water requirements—for example, rice needs abundant rainfall, while crops like pulses are more drought-tolerant. Temperature regulates plant metabolism and growth rates, with some crops flourishing in cooler climates and others, like corn, requiring warmer temperatures. Humidity plays a role in water loss through transpiration, impacting plant hydration. All these factors are interconnected, and fuzzy logic allows the system to account for their complex relationships. By integrating these essential geological and environmental inputs, the system provides more accurate and personalized crop recommendations tailored to the specific conditions of a region. This leads to improved crop yields and resource efficiency without the need for expensive, high-end testing facilities, which are often unavailable or inaccessible in rural areas (Kambalimath and Deka, 2020; Zadeh, 2023).

The study evaluates a set of 23 agricultural crops, including Rice, Wheat, Maize (Corn), Cotton, Millets, Sugarcane, Rubber, Jute, Oilseeds, Pulses, Pearl Millet (Bajra), Lentil, Groundnut, Cocoa, Tea, Coffee, Flaxseed, Coconut, Oil Palm, Clove, Black Pepper, Cardamom, and Turmeric. For defining the ranges of linguistic terms or variables associated with input parameters, the study did not divide ranges into equal intervals, such as perfectly segmenting Humidity into low, medium, and high with equal divisions. Instead, the ranges were determined based on the scientific reasoning and distribution of values in the recommendation dataset. The mean (or average) and variance (or spread of the data) informed the division of the ranges, thus ensuring the linguistic terms—such as low, medium, and high—were more contextually precise (Chen et al., 2021). Membership functions were then applied to convert the input values into their corresponding linguistic terms for each parameter.

The second stage of the fuzzy inference process involves determining the rule base of the system, which forms the foundation for decision-making. The rules are activated based on the input data. In this study, the Mamdani Fuzzy Inference System was employed due to its simplicity and widespread use in handling uncertainty and imprecision (Cherkassky, 1998). The inference rules follow an "IF-THEN' format, where linguistic variables define the conditions and outcomes (Samavat et al., 2023). This approach establishes the relationships between input and output parameters. For example, rules for crops like rice and pulses are provided in Table 2, showcasing how different inputs result in different crop recommendations. Similar rules were applied to generate a custom test dataset, leveraging the information from the recommendation dataset to validate the performance and accuracy of the recommendation system.

The final stage in the fuzzy logic process is defuzzification, which converts the fuzzy output from the inference stage into a crisp, numerical value. This stage is critical because interpreting the fuzzy outputs without defuzzification is difficult. Several methods are available for this process, including the centroid method, bisector method, mean of maxima, weighted average, weighted sum, and center of area method (Chakraverty et al., 2019). This study adopted the centroid method (also known as the center of gravity method) to defuzzify the output zone generated by the inference system. The centroid method was used to calculate the final crisp output, providing a specific crop recommendation based on the input parameters.

In essence, the fuzzy inference system operates in the following steps: First, the inputs for each variable undergo fuzzification, where the membership functions convert these values into fuzzy sets. Next, in the inference stage, the system applies predefined rules to these fuzzy sets, matching them to the inputs. Finally, the defuzzification stage uses the centroid method to generate a precise, actionable crop recommendation (Yera and Martinez, 2017). The fuzzy logic-based crop recommendation system excels in generating accurate and context-specific crop suggestions by incorporating a comprehensive range of input parameters, including soil type, geographic location, and climate conditions. Its strength lies in managing uncertainty and variability within these factors, enabling it to provide adaptive recommendations that accommodate fluctuating environmental and economic conditions. This makes the system a highly valuable tool for farmers and agricultural planners who require reliable guidance to make informed decisions in dynamic settings.

3.5. Application

This study proposes the development of an application, named "Agro Companion,' designed to be accessible across multiple devices, specifically targeting farmers and agricultural specialists. The conceptual workflow of the application is illustrated in Fig. 7. The application's interface is deliberately kept simple to ensure ease of use and quick inference processes. Given the widespread availability of smartphones, even in rural areas, the application is optimized for mobile devices to ensure broad accessibility (Montag et al., 2015). "Agro Companion' is the proposed web-based application developed using Hypertext Markup Language (HTML), Cascade Style Sheet (CSS), and JavaScript, with a Flask server handling backend process. Computationally intensive tasks, such as soil identification and crop recommendation, are performed on

Table 2
Example rules for the crops rice and pulses.

Soil	Season	pH	Rainfall	Temperature	Humidity	Nitrogen	Phosphorus	Potassium	Crop
Alluvial	Kharif	medium	high	medium	high	high	medium	medium high	*Rice*
Laterite	Kharif	medium	very high	high	high	high	medium high	high	*Rice*
Black	Kharif	medium	low	medium	low	low medium	medium	medium	*Pulses*
Red	Kharif	medium	medium	medium	medium	low medium	medium	medium	*Pulses*

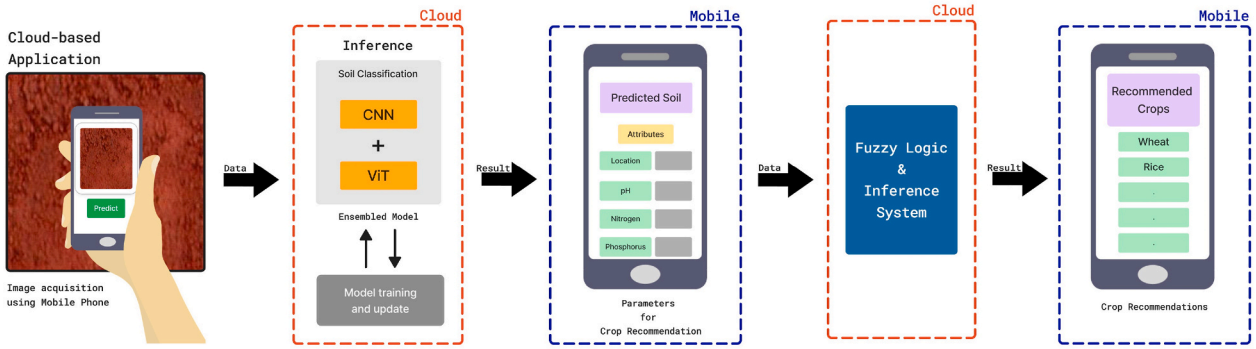


Fig. 7. Proposed smartphone based application "Agro Companion".

the cloud due to the limited processing power available on mobile devices. The application allows users to capture images of soil, which are then transmitted to the server for inference. The server uses an ensemble of Convolutional Neural Networks and Vision Transformer models to infer the soil type. Once the inference is completed, the predicted soil type is sent back to the application and displayed to the user. For crop recommendations, users input a set of parameters, which are then processed by a fuzzy logic-based inference system hosted on the cloud. The system analyzes the input and suggests suitable crops based on the provided data. The recommendations are subsequently displayed within the application (Grinberg, 2018). This proposed web-based application provides a cost-effective and user-friendly solution for soil identification and crop recommendations, offering a valuable tool for improving agricultural practices in a convenient manner.

3.6. Setup

The development and training of the GAN models, as well as other deep learning models, were conducted on a personal GPU systems equipped with an NVIDIA RTX 3080 Ti GPU, an Intel® Core™ i9-10900K central processing unit (CPU), 64 GB of random access memory (RAM), and 1 terabyte (TB) of Solid State Drive (SSD) storage. For the CycleGAN model, the official CycleGAN tutorial provided by TensorFlow was utilized as the foundation for development and validation. The original code, accessible on the TensorFlow website (https://www.tensorflow.org/tutorials/generative/cyclegan) (Abadi et al., 2016), was modified to suit the specific requirements of the project. These modifications included, but were not limited to, adjustments in the input pipeline to enhance model performance for the given use case. Deep learning models such as Convolutional Neural Networks and Vision Transformers were implemented using the PyTorch platform (Paszke et al., 2019). The fuzzy logic workflow was implemented using the Scikit-Fuzzy Python library (Warner et al., 2019), an open-source toolkit for fuzzy logic operations.

4. Results and discussion

This section presents the findings and results of the study aimed at developing a robust and accurate soil classification and crop recommendation system.

4.1. Dataset

The created dataset encompasses seven major soil types found in India: Alluvial Soil, Black Soil, Laterite Soil, Red Soil, Yellow Soil, Mountain Soil, and Arid Soil. Three major variations of this dataset were created: the original dataset collected from online sources, an augmented version of that dataset, and a CycleGAN-based augmented dataset, referred to as the CyAUG Soil Dataset. Table 3 details the size distributions across these three variations.

4.1.1. Original dataset

To ensure data quality for the soil dataset, the preprocessing steps detailed in methodology were applied. As soil images are not abundantly available online, the dataset remained relatively small. Furthermore, certain soil types are less frequently represented in online resources. For instance, Alluvial Soil and Yellow Soil were particularly underrepresented, with only 52 and 69 images, respectively, after processing the raw dataset. The final original dataset, consisting of 1189 images, was partitioned according to the methodology. Specifically, 70 % of the data (829 images) was allocated to the training set, while the validation and testing sets each received 15 % of the data, containing 182 and 178 images, respectively.

4.1.2. Augmented dataset

To address the imbalanced distribution of soil classes and the overall limited size of the original dataset, data augmentation techniques were employed. It is important to note that only the training subset was augmented, while the validation and testing sets remained unchanged to ensure consistency in model evaluation. Through the augmentation process, the size of the training set increased significantly. Originally comprising 829 images, the augmented training set grew to 3316 images—an approximate fourfold increase in size, while 15 % for both validation and testing set containing 182 images, and 178 images respectively remained same as before. This augmentation process was guided by the variations outlined in the study's methodology.

4.1.3. CyAUG dataset

While augmenting the dataset, increased the size of the training split, it did not significantly improve the overall variability and distribution of the data. When a dataset is both imbalanced and small, as was the case with the original soil dataset, conventional augmentation may fail to

Table 3
size distributions of all three variations of the soil dataset.

Soil Dataset	Alluvial Soil	Black Soil	Laterite Soil	Red Soil	Yellow Soil	Mountain Soil	Arid Soil	Total Size
Original Dataset	52	255	219	109	69	201	284	*1189*
Augmented Dataset	160	789	678	337	213	621	878	*3676*
CyAUG Dataset	837	1885	831	1430	1593	761	1076	*8413*

increase diversity meaningfully. To address these limitations, a GAN-based augmentation approach was employed, specifically leveraging CycleGAN due to the absence of paired images. Fig. 8 (A) illustrates the process of GAN-based augmentation for an imbalanced dataset. Four soil classes—Alluvial Soil, Yellow Soil, Red Soil, and Black Soil—underwent additional pruning. Images with extreme variations or overly close-up views, where the soil texture was highly distinguishable, were removed.

Four soils types were selected, and each soil class was divided into an 85 % for training set and 15 % for testing set. To improve the training of CycleGAN models and enhance precision, soil classes were further

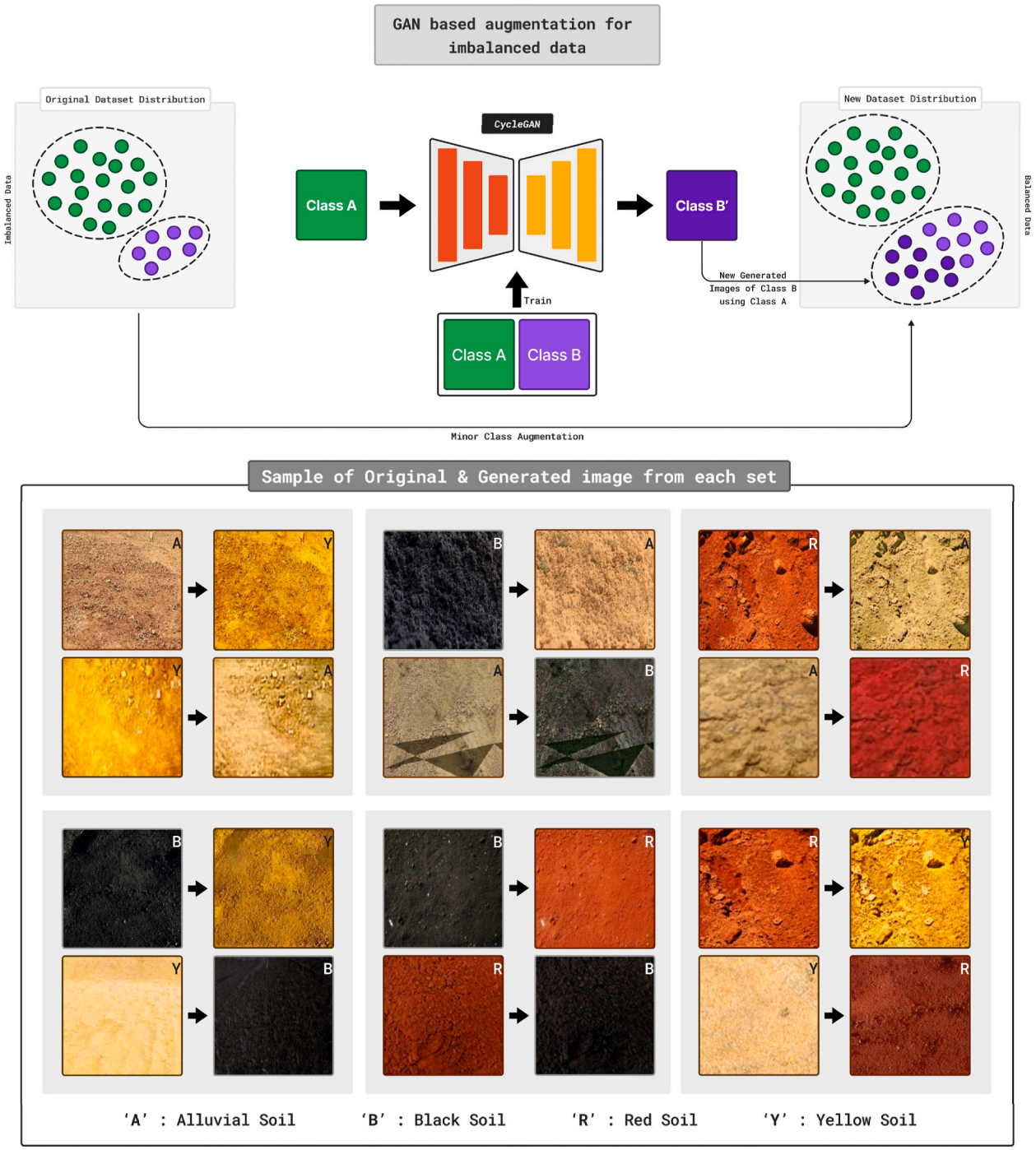


Fig. 8. Proposed GAN based augmentation (A) and samples from image translation (B).

augmented. After pruning and augmentation, the class distributions were as follows: Alluvial Soil contained 315 images, Yellow Soil had 253 images, Red Soil included 651 images, and Black Soil had 1010 images. Six CycleGAN models were then created to handle various soil class combinations: {AY: Alluvial Soil, Yellow Soil}, {BA: Black Soil, Alluvial Soil}, {RA: Red Soil, Alluvial Soil}, {BY: Black Soil, Yellow Soil}, {BR: Black Soil, Red Soil}, and {RY: Red Soil, Yellow Soil}. Each CycleGAN model was trained for 50 epochs, with training times ranging from 1.5 to 5 h depending on the size of the dataset. After training, the best-performing model variant for each set was selected based on its performance on the test set. Image translation from Class A to Class B and vice versa was performed within each pair of soil classes.

The generated images were reviewed by an agricultural expert, and those that were either translated incorrectly or of low quality were removed. Fig. 8 (B) illustrates the image translation process across all sets. After this post-processing and cleanup, the following quantities of synthetic images were generated: 642 for Alluvial Soil, 1332 for Yellow Soil, 1017 for Red Soil, and 918 for Black Soil. These newly generated synthetic images were added to the original dataset and distributed according to the previous data structure. The final augmented training set consisted of 6881 images (70 % of the total dataset), while the validation and testing sets contained 767 and 765 images, respectively (15 % each). This enhanced dataset (CyAUG dataset) effectively reduced the imbalance and increased intra-class variability.

4.2. Optimal hyperparameters

Hyperparameters, particularly the learning rate and weight decay, significantly impact the training performance of deep learning models, especially during fine-tuning. In this study, both CNN and ViT models were initially trained on the original and augmented soil datasets using a standardized set of hyperparameters, without explicit optimization or extensive tuning. These hyperparameter settings, summarized in Table 4, represents some of the commonly adopted values (Wojciuk et al., 2024; Steiner et al., 2021) and were selected through preliminary experiments. Training employed an early stopping criterion with a maximum of 30 epochs, terminating the process if the model performance did not improve for a predefined number of epochs. Furthermore, the chosen batch size of 16 was determined based on computational constraints as well as results from preliminary analyses. However, for training models on the CyAUG variant of the soil dataset, an explicit hyperparameter tuning procedure was conducted to identify optimal values for the learning rate and weight decay, aiming to maximize model performance, while the maximum number of epochs and batch size remained the same. To determine the best-performing hyperparameters, a grid search strategy was employed, focusing on the learning rate and weight decay. This approach systematically evaluates all possible combinations of these two hyperparameters to identify the optimal configuration. The ResNet50 architecture was selected for CNN-based models due to its balance between computational efficiency and high performance, while the ViT-B/16 architecture was used for Vision Transformer models. The learning rate and weight decay grid search parameters, as shown in Table 5, offered three options for each, creating a total of nine combinations. Each combination was trained for 30 epochs, with each training session taking approximately 1.5 h.

Following a thorough evaluation, the hyperparameter configuration that yielded the highest accuracy was identified. For the CNN models, the optimal combination consisted of a learning rate of 0.0001 and a weight decay of 1e-07. This configuration was subsequently applied to all CNN models trained on the CyAUG dataset. In contrast, for the ViT models, the results of the grid search indicated that the initial hyperparameter settings remained optimal, with no further adjustments needed to the learning rate and weight decay values initially presented in Table 4. The final set of hyperparameters used for CNN and ViT models for both individual model training and K-fold cross-validation on the CyAUG soil dataset remained consistent with those shown in Table 4, except for the CNN models, which adopted the updated learning rate of 0.0001 and weight decay of 1e-07. The ensemble models were trained using the CNN hyperparameters outlined in Table 4, with the exception that the maximum number of epochs was reduced to 15, as only certain layers required further training.

Table 4
Hyperparameters configurations for CNN and ViT model.

Configurations/Hyperparameters	Valuation for CNN	Valuation for ViT
Optimizer	Adam	AdamW
Batch Size	16	16
Max Epochs	30	30
Learning Rate	0.0001	1e-05
Weight Decay	1e-07	1e-07

Table 5
Hyperparameters values used for Grid Search.

Hyperparameters	Values
Learning Rate	1e-03, 1e-04, 1e-05
Weight Decay	1e-07, 1e-08, 1e-09

4.3. Transfer learning and model evaluation

Upon completing the preprocessing steps and preparing the three variants of the soil dataset, transfer learning was applied to train both Convolutional Neural Networks and Vision Transformer models. Specifically, MobileNet v3 Large, ResNet101, ResNet152, and EfficientNet v2 Large from the CNN family, and ViT-B/16 and ViT-L/16 from the Vision Transformer family, were fine-tuned following the established workflow. The models were trained on three different dataset versions: the original dataset, the augmented dataset, and the CyAUG dataset. The performance of these models was then assessed using various evaluation metrics. Given the imbalanced nature of the dataset, the primary metric for comparison was the F1-score. Additional metrics such as accuracy, precision, specificity, and sensitivity (utilized for K-fold cross-validation) were also considered to provide a comprehensive understanding of model performance. Training times varied according to dataset size and model complexity. On average, training on the original dataset took between 25 and 45 min, while the augmented dataset required 45–90 min, and the CyAUG dataset took between 1.5 and 5 h. Larger models, such as EfficientNet and the ViT variants, demanded the most computational resources and time, while MobileNet required the least. Over 30 epochs, the best-performing models for each dataset variant were selected based on their validation split performance. Fig. 9 presents a comparison of F1-score results for various deep learning models across different dataset variations, including the original dataset, augmented dataset, CyAUG dataset, and K-Fold cross-validation results (with $K = 5$) on the CyAUG dataset, using bar plots.

As demonstrated in Table 6, the performance of six deep learning models on the original dataset yielded F1-scores ranging from 85.20 % to 89.17 %. Despite the dataset's imbalance, this range reflects strong performance. Notably, the ViT-B/16 model achieved the highest F1-score (89.17 %), accuracy (89.32 %), precision (89.38 %), and specificity (98.20 %). EfficientNet v2 Large followed closely with an F1-score of 88.64 % and comparable accuracy. In contrast, MobileNet v3 Large and ResNet101 achieved similar accuracy (87.64 %), with MobileNet exhibiting a marginally higher F1-score (87.30 % vs. 87.24 %). However, ResNet152 and ViT-L/16 showed slightly lower performance, with F1-scores of 85.20 % and 85.66 %, respectively. All models demonstrated high specificity values, ranging from 97.52 % to 98.20 %, highlighting their proficiency in identifying non-target classes accurately. Precision results mirrored F1-scores closely, with ViT-B/16 again leading (89.38 %). Overall, the ViT models outperformed others due to their superior ability to handle complex and nuanced data

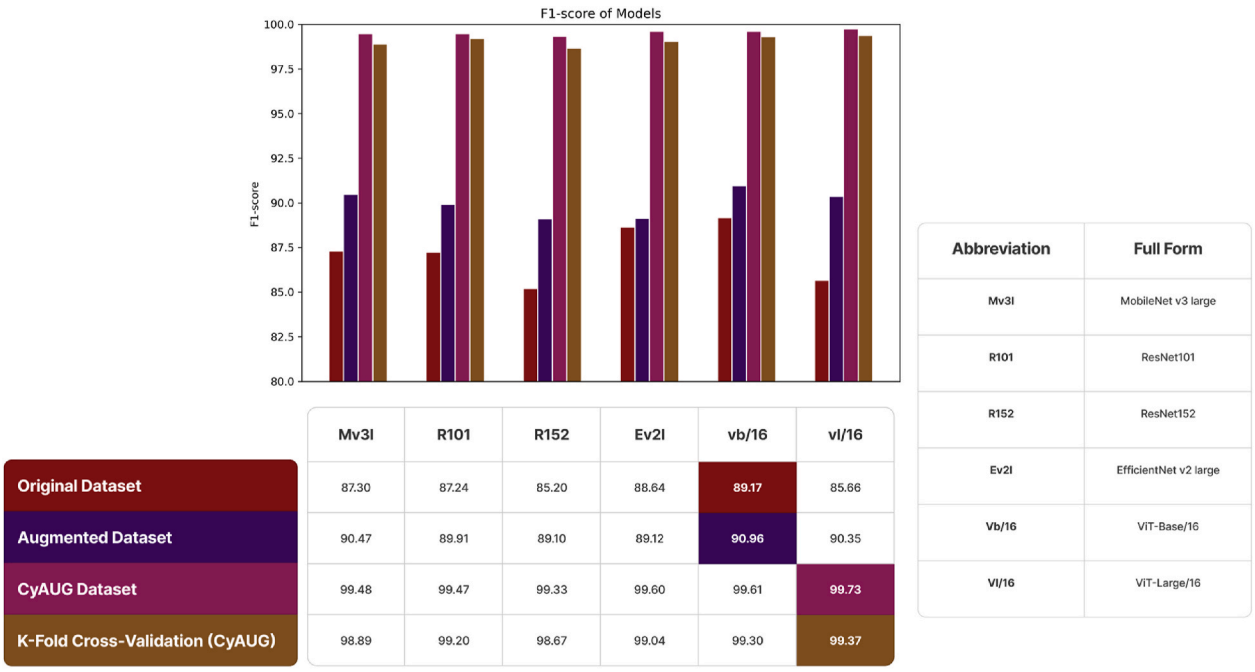


Fig. 9. Comparison of the F1-score of various deep learning models across different dataset variations.

Table 6
Deep learning models' soil classification performance on original dataset.

Model	Accuracy (%)	F1-score (%)	Precision (%)	Specificity (%)
MobileNet v3 large	87.64	87.30	87.73	97.91
ResNet101	87.64	87.24	87.75	97.86
ResNet152	85.39	85.20	85.66	97.52
EfficientNet v2 large	**89.32**	88.64	88.38	97.91
ViT-B/16	**89.32**	**89.17**	**89.38**	**98.20**
ViT-L/16	85.95	85.66	85.79	97.61

representations, making them particularly well-suited for soil classification task. Fig. 10 (A) presents the training and loss curves for models trained on the original dataset.

When data augmentation techniques were applied, significant improvements were observed across all models, as shown in Table 7. Data augmentation slightly alleviated the imbalance issue and improved model generalization. F1-scores for all models increased, ranging from 89.10 % to 90.96 %. Once again, the ViT-B/16 model achieved the highest F1-score (90.96 %), along with the highest accuracy, precision, and specificity, further validating its robustness in learning complex patterns. MobileNet v3 Large also exhibited a notable increase in performance, reaching an F1-score of 90.47 %, significantly surpassing its performance on the original dataset. ResNet101 followed closely, with an F1-score of 89.91 %. On the other hand, EfficientNet v2 Large showed only marginal improvement, with an F1-score of 89.12 %. Meanwhile, ResNet152 continued to lag behind, with an F1-score of 89.10 %. Specificity values remained consistently high, exceeding 98.15 % across all models, emphasizing the models' ability to minimize false positives. Precision scores also improved, with ViT-B/16 (91.16 %) and ResNet101 (90.98 %) standing out. Though there were performance gains across the board, the improvements from the original dataset to the augmented dataset were not drastic. As observed in Fig. 10 (B), models trained on augmented datasets not only learned faster but also began with higher initial accuracy compared to those trained on the original dataset. A similar trend can be observed in the loss plots, indicating that data augmentation had a positive effect on both learning speed and final performance outcomes.

Table 7
Deep learning models' soil classification performance on augmented dataset.

Model	Accuracy (%)	F1-score (%)	Precision (%)	Specificity (%)
MobileNet v3 large	90.44	90.47	90.59	98.34
ResNet101	90.44	89.91	90.98	98.32
ResNet152	89.88	89.10	89.58	98.23
EfficientNet v2 large	89.32	89.12	89.16	98.16
ViT-B/16	**91.01**	**90.96**	**91.16**	**98.49**
ViT-L/16	90.44	90.35	90.77	98.35

The results in Table 8 underscore the significant performance improvements achieved using the CyAUG dataset, which represents the final, optimized version incorporating targeted CycleGAN-based augmentation to address class imbalance and enhance feature diversity. The F1-scores observed in this table are remarkably high, ranging from 99.33 % to 99.73 %, reflecting a substantial improvement over the performance on the augmented dataset. Leading the models is ViT-L/16, with an exceptional F1-score of 99.73 %, demonstrating its superior classification capability compared to its earlier results. Following closely, EfficientNet v2 Large and ViT-B/16 exhibit nearly identical F1-scores of 99.60 % and 99.61 %, respectively, alongside outstanding performance across other metrics, including precision and specificity, both of which exceed 99.9 %. MobileNet v3 Large and ResNet101, though slightly behind the top performers, still deliver remarkable results, with F1-scores of 99.48 % and 99.47 %, respectively. ResNet152, while exhibiting the lowest performance among all models, achieves a still-impressive F1-score of 99.33 %, along with robust performance across all other metrics, maintaining values above 99 %. The models also achieve near-perfect accuracy values, ranging from 99.34 % to 99.73 %, indicating that the vast majority of predictions are correct. Specificity scores are virtually flawless, exceeding 99.88 %, while precision values, ranging from 99.35 % to 99.74 %, align with the high F1-scores, highlighting the models' ability to minimize false positives while maintaining a high rate of true positive predictions. Fig. 10 (C) illustrates the training accuracy and loss plots for models trained on the

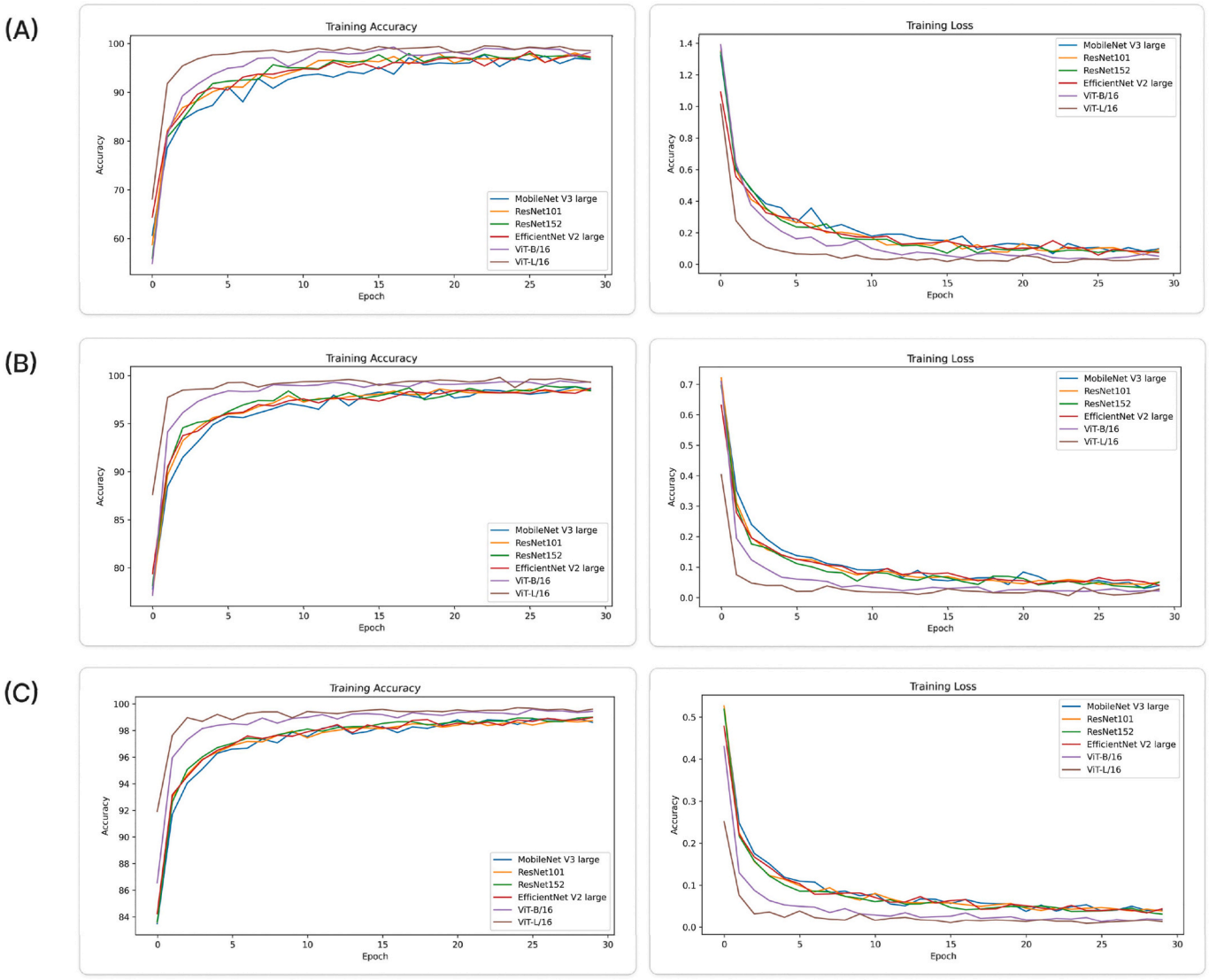


Fig. 10. Training accuracy and loss plot of deep learning models over the original dataset (A), augmented dataset (B), and CyAUG dataset (C).

Table 8
Deep learning models' soil classification performance on CyAUG dataset.

Model	Accuracy (%)	F1-score (%)	Precision (%)	Specificity (%)
MobileNet v3 large	99.47	99.48	99.49	99.92
ResNet101	99.47	99.47	99.49	99.91
ResNet152	99.34	99.33	99.35	99.88
EfficientNet v2 large	99.60	99.60	99.61	99.93
ViT-B/16	99.60	99.61	99.61	99.93
ViT-L/16	**99.73**	**99.73**	**99.74**	**99.95**

CyAUG dataset, revealing that these models converged significantly faster compared to those trained on other dataset variants. The ViT models, in particular, reached optimal accuracy much quicker than CNN-based models, a trend also reflected in their loss curves. Overall, all models converged to optimal accuracy more efficiently on the CyAUG dataset than on other soil dataset variants, showing effectiveness of the CycleGAN-based data augmentation strategy.

Table 9 presents the results of K-fold cross-validation (K = 5) on the CyAUG dataset to evaluate the stability and generalization capability of the models. All models achieve near-perfect results, with F1-scores exceeding 98.6 %. The ViT-L/16 model remains the top performer across all folds, with the highest F1-score (99.37 %), accuracy (99.71 %), precision (99.53 %), and specificity (99.95 %). ViT-B/16 and EfficientNet v2 Large also show strong results, with F1-scores of 99.30 % and 99.04 %, respectively. Although ResNet101 and MobileNet v3 Large present slightly lower F1-scores (99.20 % and 98.89 %, respectively), their performance remains highly competitive. ResNet152, with an F1-score of 98.67 %, lags behind the other models but still delivers robust

Table 9
Deep learning models' soil classification performance over 5-fold cross validation on CyAUG dataset.

Model	Accuracy (%)	F1-score (%)	Precision (%)	Sensitivity (%)	Specificity (%)
MobileNet v3 large	99.47	98.89	98.90	98.92	99.91
ResNet101	99.50	99.20	99.17	**99.26**	**99.95**
ResNet152	99.24	98.67	98.95	98.44	99.88
EfficientNet v2 large	99.55	99.04	99.20	98.92	99.91
ViT-B/16	99.66	99.30	99.44	99.17	99.91
ViT-L/16	**99.71**	**99.37**	**99.53**	99.24	**99.95**

performance. Notably, ResNet101 achieved the highest sensitivity score (99.26 %), reflecting its superior ability to correctly identify true positives. Accuracy and specificity remain near-perfect across all models, while precision values are also exceptionally high, with ViT-L/16 achieving the top score at 99.53 %. Overall, these results confirm that Vision Transformer models, especially ViT-L/16, consistently outperform other architectures across all datasets and evaluation setups. The cross-validation results further validate the generalization capability of the models, with minimal performance variation across folds. Additionally, lightweight models like MobileNet v3 Large demonstrate competitive performance despite their smaller architecture, offering an excellent balance between model size and performance.

In conclusion, the experimental findings suggest that ViT and EfficientNet models are the most suitable for soil classification tasks, particularly when trained on the CyAUG dataset, as they offer a compelling combination of high accuracy and computational efficiency. These results also emphasize the critical role of proper data augmentation and large-scale datasets in improving classification performance, especially when dealing with imbalanced and small datasets.

4.4. Ensembled models

Although the individual deep learning models achieved near-perfect accuracy and F1-scores on the CyAUG dataset, the introduction of ensemble models offers a more robust and reliable performance. The training time for the ensemble models ranged from 30 to 90 min, with the duration increasing based on the number and type of architectures involved. However, the overall training time remained comparatively low, as only the newly added layers in the ensemble architecture were fine-tuned, while the fine-tuned CNN and ViT model layers remained frozen.

The performance of eight ensemble models is summarized in Table 10. Most ensembles achieved near-perfect F1-scores, ranging from 99.48 % to 100 %. The most significant performance was observed in the **EffL-VL** ensemble, which combines EfficientNet v2 Large and ViT-L/16. This model attained a perfect F1-score of 100 %, alongside 100 % accuracy, precision, and specificity, demonstrating its exceptional reliability, with no misclassifications across the dataset. Similarly, ensembles such as **R152-VL**, **R152-EffL-VL**, and **Mob-R152-EffL-VL** showed near-perfect F1-scores of 99.86 %, with equivalent results in accuracy (99.86 %), precision (99.87 %), and specificity (99.98 %). Similarly, **Mob-R101-EffL-VB**, with an F1-score of 99.73 %, performed well but slightly lower than the EffL-VL, R152-VL, R152-EffL-VL, and Mob-R152-EffL-VL ensembles. These findings highlight that combining deeper architectures, like ResNets, with Vision Transformers or EfficientNets, harnesses the strengths of both CNNs and ViTs more strongly.

Other ensemble models, such as **Mob-VB** (MobileNet v3 Large + ViT-B/16), **R101-VB** (ResNet101 + ViT-B/16), and **Mob-R101-VB** (MobileNet v3 Large + ResNet101 + ViT-B/16), also exhibited strong performance, with F1-scores of 99.74 %, 99.48 %, and 99.61 %, respectively, alongside high accuracy, precision, and specificity. Although R101-VB had the lowest performance among the ensemble models, the inclusion of MobileNet architecture in the R101-VB ensemble contributed to an overall improvement in performance. These combinations of lightweight and deeper models ensures that the ensemble benefits from both computational efficiency and high representational power. Additionally, lightweight ensembled model Mob-VB showed very high performance despite being the smallest ensembled model, offering an excellent balance between model size, speed and performance.

Overall, the ensemble models exhibited greater robustness and consistency compared to the individual models. Ensemble learning capitalizes on the complementary strengths of diverse architectures, enabling models to correct each other's weaknesses. This results in more reliable outcomes, providing an additional layer of precision and improved generalization for complex tasks such as soil classification.

Table 10
Ensembled CNN and ViT models performance.

Model	Accuracy (%)	F1-score (%)	Precision (%)	Specificity (%)
Mob-VB	99.73	99.74	99.75	99.96
R101-VB	99.47	99.48	99.48	99.91
R152-VL	99.86	99.86	99.87	99.98
EffL-VL	**100**	**100**	**100**	**100**
Mob-R101-VB	99.60	99.61	99.62	99.93
R152-EffL-VL	99.86	99.86	99.87	99.98
Mob-R101-EffL-VB	99.73	99.73	99.73	99.95
Mob-R152-EffL-VL	99.86	99.86	99.87	99.98

4.5. Recommendation system

The crop recommendation system developed in this study is based on a fuzzy inference system (FIS), which functions through three core stages: fuzzification, inference, and defuzzification. This system leverages nine input parameters, categorized into two groups: geological factors (Soil, pH, Nitrogen, Phosphorus, and Potassium content) and environmental factors (Season, Rainfall, Temperature, and Humidity). To model these inputs, triangular (trimf) and trapezoidal (trapmf) membership functions were employed. Each input parameter is mapped to a corresponding set of linguistic terms, such as *Low*, *Medium*, or *High*, that represent distinct ranges of possible values. The details of all parameters, their descriptions, and their associated ranges are provided in Table 11. This table quantifies each parameter and specifies its possible values, serving as the basis for both creating the membership functions and establishing the fuzzy rule base.

For discrete parameters (e.g., soil type and season), numerical labels that simplify membership mapping are assigned. For instance, soil types Alluvial, Black, Red, Yellow, Laterite, Arid, and Mountain are encoded as 0, 1, 2, 3, 4, 5, and 6, respectively. Similarly, the three season categories, Rabi, Kharif, and Year-Long are encoded as 0, 1, and 2, respectively. For continuous parameters such as pH, rainfall, temperature, and humidity, Table 11 defines the value ranges over which membership functions are formed. These ranges are then divided into linguistic terms (e.g., *Low*, *Medium*, *High*) to facilitate fuzzy inference, as illustrated in Fig. 11. Soil nutrient levels (nitrogen, phosphorus, potassium) are also discretized into five categories (0, 1, 2, 3, 4), each corresponding to a specific linguistic label. For example, a value of 0 indicates *Low* content, 1 indicates *Low-Medium*, 2 represents *Medium*, 3 corresponds to *Medium-High*, and 4 represents *High* content. This discretization strategy extends to crop types as well, where unique numerical labels enable easy mapping to membership functions.

The advantage of using these discrete numeric encodings lies in the simplicity of constructing fuzzy rules. Since the fuzzy inference system operates on numerical inputs, mapping qualitative descriptors (e.g., *Low*, *Medium*, *High*) to numeric values streamlines both membership function creation and rule formulation. Table 12 lists the complete set of

Table 11
Inference System's parameters, their description, and values.

Parameters Description	Parameters	Discrete Values/Ranges
The type of soil	Soil	0, 1, 2, 3, 4, 5, 6
Season	Season	0, 1, 2
pH	pH	0–10
Average rainfall (in cm)	Rainfall	0–500
Average temperature (in °C)	Temperature	0–50
Average humidity (in %)	Humidity	0–100
Nitrogen content in soil	Nitrogen	0, 1, 2, 3, 4
Phosphorus content in soil	Phosphorus	0, 1, 2, 3, 4
Potassium content in soil	Potassium	0, 1, 2, 3, 4

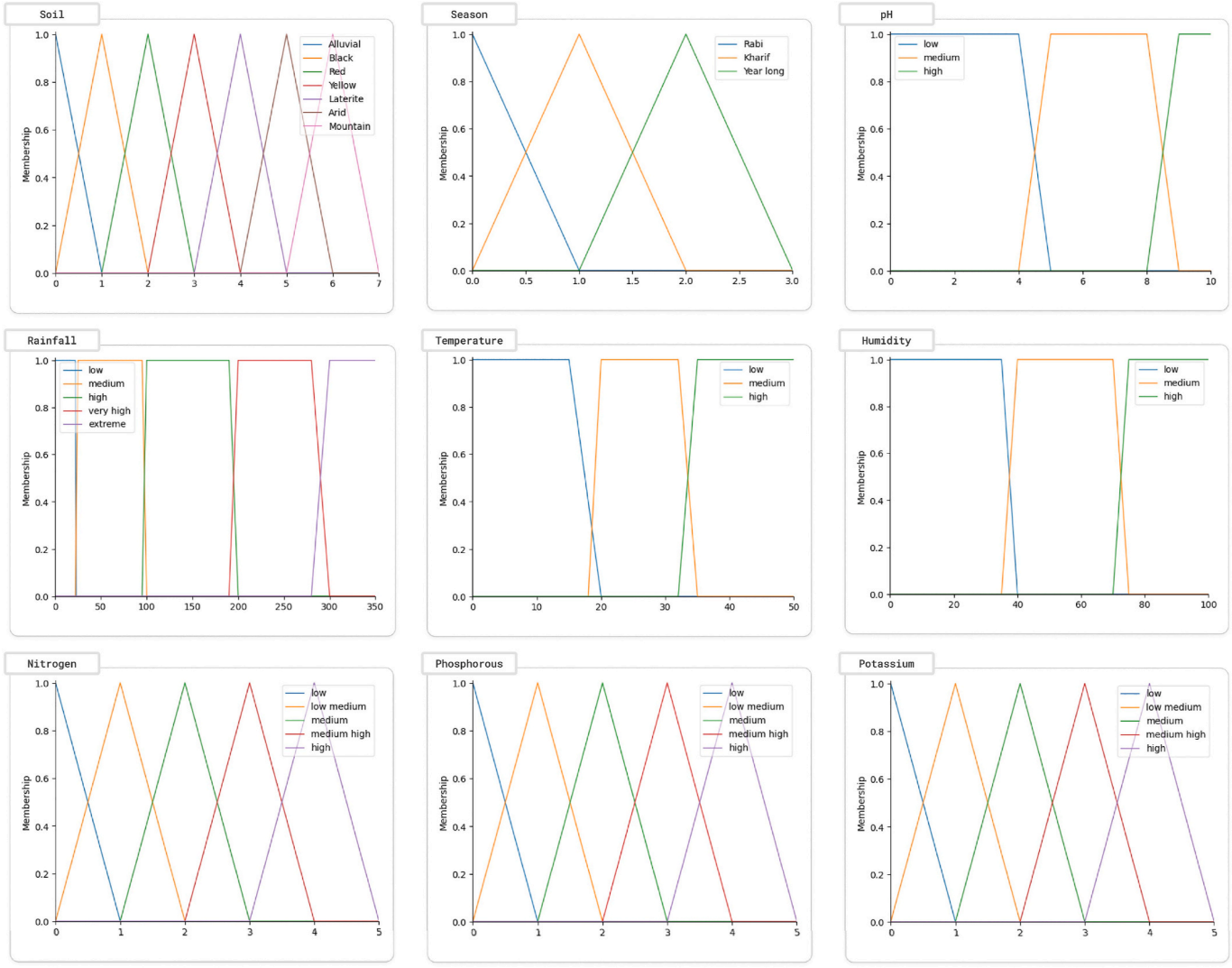


Fig. 11. Graphical representation of the variables' membership functions.

Table 12
Inference System's parameters, their linguistic terms and membership functions.

Parameters	Linguistic Terms/Variables	Membership function
Soil	Alluvial, Black, Laterite, Red, Yellow, Mountain, Arid	trimf
Season	Rabi, Kharif, Year long	trimf
pH	low, medium, high	trapmf
Rainfall	low, medium, high, very high, extreme	trapmf
Temperature	low, medium, high	trapmf
Humidity	low, medium, high	trapmf
Nitrogen	low, low-medium, medium, medium-high, high	trimf
Phosphorus	low, low-medium, medium, medium-high, high	trimf
Potassium	low, low-medium, medium, medium-high, high	trimf
Crops	Rice, Wheat, Maize (Corn), Cotton, Millets, Sugarcane, Rubber, Jute, Oilseeds, Pulses, Pearl Millet (Bajra), Lentil, Groundnut, Cocoa, Tea, Coffee, Flaxseed, Coconut, Oil palm, Clove, Black pepper, Cardamom, Turmeric	trimf

linguistic terms and their corresponding membership functions. Fig. 11 further visualizes these membership functions, highlighting how the discrete encodings in Table 11 translate into the fuzzy sets used in subsequent rule-based reasoning. This structured mapping process ensures consistency and interpretability. By systematically associating each parameter with well-defined linguistic and numerical ranges, the framework helps supports robust rule generation, clear parameter boundaries, and improved transparency in the decision-making process.

A Mamdani Fuzzy Inference System was employed to derive the crop recommendations using a set of "IF-THEN' inference rules. The system is capable of recommending 23 different crops, with 270 hand-crafted inference rules created based on the recommendation dataset. These rules consider the nine specified input parameters and generate a crop recommendation from the set of 23 agricultural crops. To account for practical limitations, Nitrogen, Phosphorus, and Potassium were treated as optional inputs. This approach addresses the challenge that obtaining accurate information on soil nutrients is not always feasible. As a result, the system is designed to operate effectively even when these nutrient values are unavailable.

Defuzzification was conducted using the centroid method, which transforms the fuzzy output into a crisp numerical value that corresponds to a specific crop recommendation. This ensures that the recommendations are accurately tailored to the provided input parameters. The system was evaluated using a custom-developed rule-based test dataset, derived from the recommendation dataset. The recommendation system achieved a 94 % accuracy rate, aligning closely with the ground truth labels in the test dataset. Additionally, it generated supplementary crop suggestions, which were deemed plausible based on the input geological and environmental factors. These extra

recommendations further demonstrated the adaptive capabilities of the system, validating its robustness and flexibility in diverse conditions. The system offers a practical and efficient solution for generating precise crop recommendations based on both geological and environmental factors, without the need for costly testing facilities.

4.6. Application

The proposed web-based application, "Agro Companion," was developed using HTML, CSS, and JavaScript for the frontend, with a Flask-based server for inference and crop recommendations. This application, built upon the framework proposed in the Methodology section, performs identification, inference, and various calculations on the cloud. The application is accessible on both mobile and desktop platforms, ensuring broad usability. Fig. 12 illustrates the key components of the working application. The homepage of "Agro Companion' is depicted in Fig. 12 (A), offering a user-friendly interface. The soil detection functionality, illustrated in Fig. 12 (B), enables users to upload images of soil samples, which are subsequently classified into one of seven predefined soil categories. The identified soil class is subsequently displayed to the user. Classification is performed using an optimal ensemble deep learning model, selected based on the trade-off between prediction accuracy and computational efficiency. Specifically, the ensemble model Mob-VB, combining MobileNet v3 Large and Vision Transformer (ViT-B/16), was chosen due to its superior performance despite being the most lightweight ensemble tested. Mob-VB demonstrated exceptional performance metrics, effectively balancing computational complexity, inference speed, and classification accuracy. Such a balance is particularly beneficial for deployment in resource-constrained environments, ensuring usability without compromising predictive reliability. For crop recommendation, the framework is demonstrated in Fig. 12 (C). Users can input various parameters,

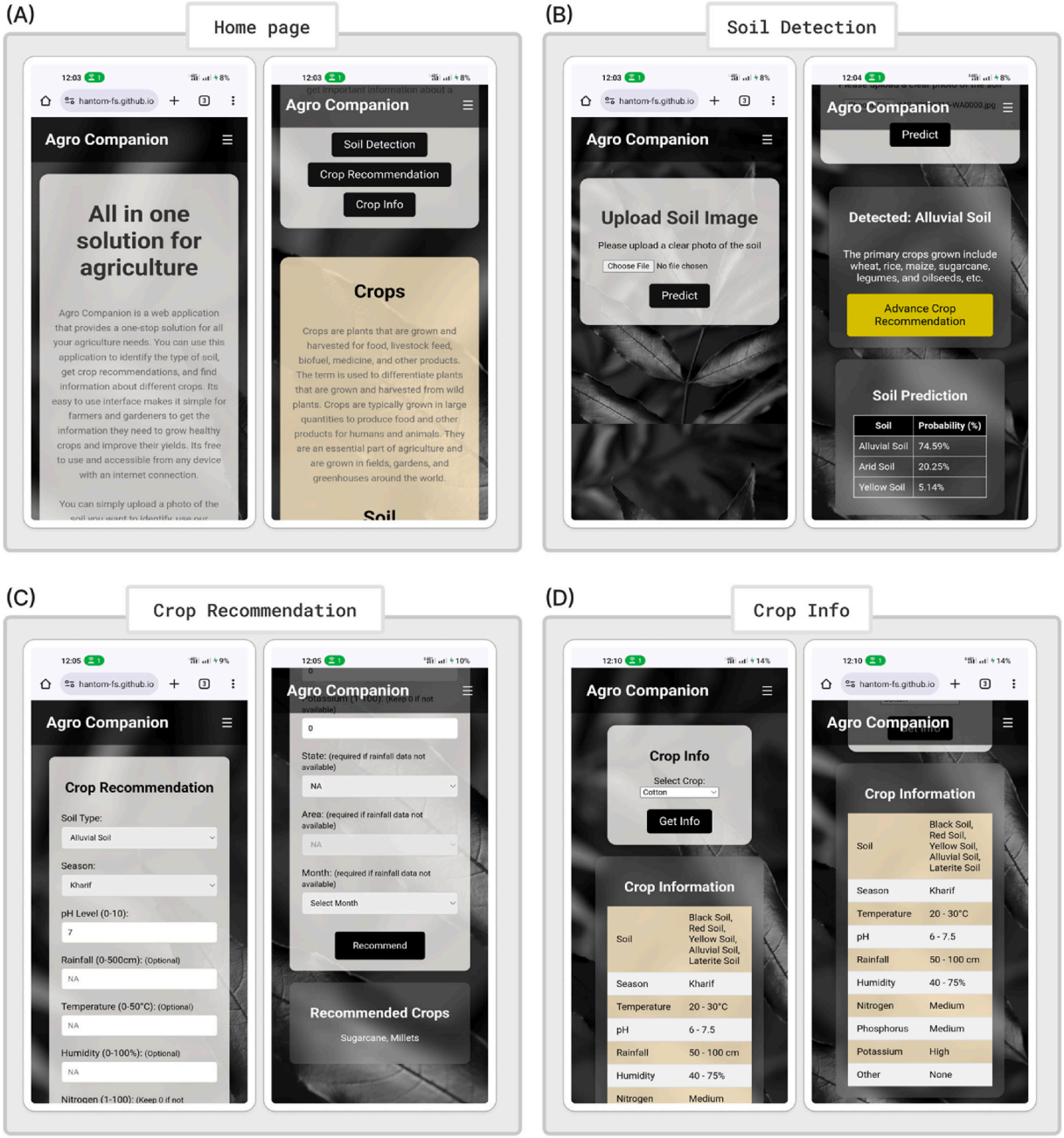


Fig. 12. "Agro Companion", a web based application for soil identification and crop recommendation.

including soil type (identified via the soil detection feature), season, pH, rainfall, temperature, humidity, nitrogen, phosphorus, and potassium levels. If temperature and humidity data are unavailable, the user can leave these fields blank, and the application will use the device's Global Positioning System (GPS) to obtain latitude and longitude data, which is then utilized to fetch average humidity and rainfall forecasts for the next month via an integrated weather Application Programming Interface (API). Similarly, if rainfall data is not readily available, the user can select their location (Indian state and region) and the relevant month, and the application will retrieve rainfall data from the recommendation dataset, which is sourced from the Indian Meteorological Department's website. The application also includes a crop information feature, as shown in Fig. 12 (D). Users can select from 23 different crop types, and the application provides detailed information on optimal growing conditions and other essential data for each selected crop. By integrating real-time weather data, leveraging deep learning models for soil classification, and fuzzy logic for crop recommendation, "Agro Companion' offers a comprehensive tool for farmers to make informed decisions about crop cultivation, all within an easy-to-use digital platform.

5. Conclusion

This research presents Soil-ViT-CNN, a novel AI-based framework that ensembles Vision Transformers and Convolutional Neural Networks for highly accurate soil classification, integrated with a fuzzy logic-based crop recommendation system. By leveraging CycleGAN-based data augmentation, the dataset was expanded to 8413 images, improving model robustness and generalization. The ensemble approach significantly outperformed individual models, achieving up to 100 % classification accuracy, precision, and specificity. Notably, ViT-L/16 and EfficientNet v2 demonstrated outstanding performance with 99.73 % and 99.60 % accuracy, respectively, on the CyAUG dataset. The combination of CNNs and ViTs effectively captures both local and global features, making it an optimal solution for soil classification. Beyond classification, the study introduces a fuzzy logic-driven crop recommendation system, incorporating key environmental and agricultural factors such as soil nutrients (N, P, K), pH levels, and climatic conditions. This comprehensive decision-making framework provides context-aware recommendations, addressing a crucial gap in existing agricultural advisory systems. By offering a mobile/web-based platform, the system enhances accessibility for farmers and agricultural stakeholders, allowing them to make data-driven decisions, optimize resource utilization, and reduce land degradation caused by suboptimal crop selection.

While the Soil-ViT-CNN framework demonstrates exceptional performance, several avenues for improvement can enhance its real-world applicability and scalability like Dataset Expansion & Multispectral Imaging, Enhanced Crop Recommendation Using Advanced AI Techniques, IoT-Enabled Real-Time Soil Monitoring, Explainable AI (XAI) for Improved Adoption & Trust and Scalability & Global Implementation. Overall, this research lays a strong foundation for AI-driven smart agriculture, demonstrating how deep learning and fuzzy logic can transform traditional soil classification and crop selection. By continuing to enhance dataset diversity, improve recommendation accuracy, and integrate real-time monitoring, Soil-ViT-CNN has the potential to revolutionize modern farming, increase crop yields, and promote sustainable agricultural practices globally.

CRediT authorship contribution statement

Farhan Sheth: Methodology, Investigation, Conceptualization. **Priya Mathur:** Writing – review & editing, Supervision, Formal analysis. **Amit Kumar Gupta:** Writing – original draft, Visualization, Methodology, Data curation, Conceptualization. **Sandeep Chaurasia:** Visualization, Supervision, Conceptualization.

Code Availability

The core modules, which include scripts for data preprocessing, model training, and the implementation of the fuzzy recommendation engine, are hosted on GitHub. The repository can be accessed at https://github.com/Phantom-fs/Agro-Companion-Modules, offering comprehensive resources for both model development and system evaluation. The web-based interface that allows users to interact with the AI model for soil identification and retrieving personalized crop recommendations is also publicly available. The website, its code, alongside the necessary API integrations, can be found at https://github.com/Phantom-fs/Agro-Companion. All repositories include the required documentation for installation, setup, and customization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The curated dataset utilized for soil classification has been made publicly available and can be accessed at https://github.com/Phantom-fs/Soil-Classification-Dataset.

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